Project Report

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Definition of HRC & MoC modelling

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Abstract

This study addresses the homogeneous risk classification (HRC) and the modeling of the margin of conservatism (MoC) in the context of credit risk. The objective is to explore existing methods and propose a methodological framework for constructing homogeneous risk classes and estimating the margins of conservatism.

The methodology is based on the creation of HRC from the provided scores using various approaches: the CART algorithm (Gini criterion and Shannon entropy), the weight of evidence (WOE) approach, the quantile-based approach, and dynamic programming with the BRKGA algorithm. The homogeneity and stability of the classes are assessed using statistical tests (Chi-square, Welch, Kolmogorov-Smirnov, ANOVA). The estimation of margins of conservatism relies on parametric methods, bootstrapping, and a Bayesian approach, with a comparison of the results obtained for each method.

The results show that the CART approach is the most suitable for HRC creation due to its simplicity and performance. The bootstrap method appears to be the most reliable and conservative for MoC estimation. However, challenges remain, particularly concerning the temporal stability of risk classes and the quality of the initial score model.

Résumé

Cette étude traite de la classification homogène des risques (HRC) et de la modélisation de la marge de prudence (MoC) dans le contexte du risque de crédit. L'objectif est d'explorer les méthodes existantes et de proposer une approche méthodologique pour la construction des classes de risque homogènes et l'estimation des marges de prudence.

La méthodologie adoptée repose sur la création de HRC à partir des scores fournis en utilisant différentes approches : l'algorithme CART (critère de Gini et entropie de Shannon), l'approche par poids de l'évidence (WOE), l'approche par quantiles et la programmation dynamique avec l'algorithme BRKGA. L'homogénéité et la stabilité des classes sont évaluées par des tests statistiques (Chi-deux, Welch, Kolmogorov-Smirnov, ANOVA). L'estimation des marges de prudence repose sur des méthodes paramétriques, le bootstrapping et une approche bayésienne, avec une comparaison des résultats obtenus pour chaque méthode.

Les résultats montrent que l'approche CART est la plus adaptée pour la création des HRC en raison de sa simplicité et de sa performance. La méthode bootstrap apparaît comme la plus fiable et conservatrice pour l'estimation des marges de prudence. Cependant, des défis persistent, notamment en matière de stabilité temporelle des classes de risque et de qualité du modèle de score initial.

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Introduction

Banks and financial institutions face a number of risks in the course of their business, which they seek to control. This need goes beyond an internal necessity, as the national and even global economic system operates under the constraint of a fragility in the sector's mode of operation. This can be seen in the presence of regulators who ensure that the standards laid down are applied. These standards provide a roadmap for dealing with risk, particularly credit risk, which is the focus of this study.

Defined as the possibility that a borrower may be unable to repay his or her loans, credit risk is subject to ongoing monitoring, marked by the continuous improvement of quantification approaches. One of the most significant of these is the introduction of the internal ratings based approach (IRB) under Basel 2 in 2004. However, the European Banking Authority's (EBA) report on IRB modeling practices highlights the wide diversity of approaches used by institutions, which justifies the harmonization targeted by the EBA's guidelines on Probability of Default (PD), Loss Given Default (LGD) and the treatment of defaulting exposures. In addition to the divergences noted in their investigation, the authority also stresses the importance of a margin of conservatism (MoC) to take account of estimation errors. Indeed, the investigations revealed that 20 % of default probability models do not include a margin of conservatism (European Banking Authority 2017).

At the end of this survey, we found that the approach to estimating Basel parameters and the rigor with which uncertainties are controlled were both diversified. All this leads to unjustified variability in RWA. This raises the question of which internal approaches would best model the probability of default through a homogeneous definition of risk classes, while providing methodically justifiable margins for conservatism. The answer to this lies in achieving two objectives: the first is to develop homogeneous classes of risk that best combine individual scores to quantify the probability of default by homogeneous groups, and the second is to reflect the statistical uncertainty affecting final estimates of the probability of default. To achieve this objective, a preliminary review will be carried out in order to master the existing situation. This will be followed by a presentation of the methodology used and the results obtained. Finally, we'll end with a discussion and conclusion.

1 Exploration of the existing HRC and MoC credit risk

The creation of risk classes and the definition of conservation margins is not a recent requirement concerning credit risk. As such, numerous related studies have been conducted in this regard. This section aims to provide a non-exhaustive review in this context.

1.1 Validation results of internal models: focus on the Probability of Default (PD)

The European Central Bank (ECB) provides a detailed framework for reporting the validation results of internal models, focusing on all key risk parameters, including Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). As the primary focus here is on PD, the following outlines the specific instructions for validating PD models, as described in the ECB's guidelines (European Central Bank, Banking Supervision 2019). The ECB stresses the importance of applying both qualitative and quantitative validation tools to assess the performance of PD models. These tools aim to evaluate the model's predictive ability, discriminatory power, and stability over time. Institutions must report their findings in a structured manner, addressing each of these aspects.

From a qualitative perspective, the ECB mandates that institutions evaluate the process by which ratings are assigned and how defaults are identified. The validation should assess whether the data used for the model is appropriate and up-to-date, and whether the rating process itself is consistent and transparent. Specifically, the ECB requests that institutions report the size of the portfolio, the number of clients, and the frequency of overrides and technical defaults, among other metrics. These qualitative statistics are crucial for understanding potential weaknesses in the rating process and the reliability of the PD model.

On the quantitative side, the ECB requires institutions to assess the calibration, discriminatory power, and stability of the PD model. Calibration refers to the model's ability to accurately predict default events, and this is often evaluated using statistical tests such as Jeffreys' test, which compares predicted defaults with observed defaults. The discriminatory power of the model is evaluated using the Area Under the Curve (AUC), which measures the model's ability to rank clients by their likelihood of default. Finally, the stability of the model is assessed over time, using tools like migration matrices and concentration indices to monitor whether the model maintains its predictive performance and consistency in different time periods.

Institutions are also expected to report the results of these validation tests, including the p-values, test statistics, and any observations regarding the model's performance. This transparency is essential to ensure that the model meets the ECB's standards for risk modeling and remains effective over time. These different instructions align with the expectations of the Basel Committee regarding the same issue of internal model validation (Banking Supervision 2005). A reminder of the need for comparability with external ratings is also emphasized. Ultimately, it is important to note that there is no universal method for validating PDs, and the choice of appropriate methods depends on the type of rating system and the available data.

1.2 Review of the creation of homogeneous risk classes

In the context of credit risk, Homogeneous Risk Class (HRC) modeling involves grouping borrowers with similar risk characteristics in order to more accurately estimate the probability of default. This approach is crucial for effective credit risk management and capital allocation. Within this framework, borrowers are divided into groups based on factors such as credit scores, financial ratios and business sectors. Within each class, borrowers are assumed to have similar default probabilities, enabling a more accurate assessment of risk. The asymptotic Vasicek single risk factor model is often used in this context, dividing default risk into systematic and entity-specific components (Yang 2014). This approach enables financial institutions to model systematic risk using multifactor models, and entity-specific risk using probit or logistic models.

Furthermore, it is interesting to note that some studies suggest that traditional HRC models may not fully capture the complexities of credit risk. For example, the long-range, finite-size Ising model has been proposed as an alternative for describing homogeneous credit portfolios, particularly when default correlations between borrowers are taken into account (Molins and Vives 2004). This model indicates that increasing correlations can lead to a sudden increase in risk, a feature not usually captured by standard credit risk models. Beyond this state of the art, ongoing research suggests that more sophisticated models may be needed to fully capture the nuances of credit risk, particularly in highly correlated portfolios.

1.2.1 Expectations for homogeneous risk classes

As said earlier, the main goal of creating these homogeneous risk classes is to group exposures or obligors with similar risk characteristics into the same class or pool. This ensures that the risk parameters (PD, LGD) calculated for each class are representative of the risk level of all the elements within it. Homogeneity is evaluated based on the similarity of loss characteristics of the obligors and transactions included in each class or pool.

Key aspects of homogeneous risk class creation include (EBA 2017; EBA 2016):

- Risk drivers: Risk classes are created using relevant risk drivers that help differentiate the risks between obligors or exposures. These include:
  - Obligor characteristics: financial information, qualitative information, sectoral risk, country risk, and parent company support for businesses and institutions. For retail exposures, this includes income, behavior, and socio-demographic data.
  - Transaction characteristics: product type, collateral type, loan maturity, loan-to-value ratio.
  - Information on defaults, internal data, or external sources such as credit rating agencies.
  - Exposure: If exposure value is a significant risk factor, it should be used for segregating or differentiating the risk of LGD.
  - Number of loans of the same type granted to the same obligor.
- Assignment process: Exposures must be assigned to classes consistently and unambiguously, following the correct sequence. Exposures are first classified based on transaction characteristics, then by those of the obligor. Assignment criteria should be sufficiently detailed to ensure consistent application and clear understanding by the staff. All exposures to the same obligor must be assigned to the same class, including across different sectors or legal entities of the group. Overrides must be justified and not indicate significant weaknesses in the rating model.
- Number of classes and distribution: The number of classes should be sufficient to allow meaningful risk differentiation and loss quantification. The concentration of exposures or obligors in a single class should not be excessive unless justified by empirical evidence of risk homogeneity. There should not be too few exposures or obligors in a single class.

- Homogeneity and default risk: Grades must be defined in such a way that each obligor within a grade or pool has a reasonably similar default risk. Significant overlap in default risk distributions between grades or pools should be avoided.

- Statistical models and mechanics: When using statistical models, it is essential to ensure that the models are based on adequate data, considering their quality and representativeness. The institution must be aware of the limitations of the models and ensure that they are used appropriately.

- Monitoring and adjustments: Institutions must monitor the distribution of exposures and obligors within the different classes. In case of insufficient data, an increased margin of conservatism should be adopted when calculating risk weights.

- Use of data: The data used to create homogeneous risk classes must be representative of the applicable portfolio. Institutions must analyze the representativeness of the data, including in terms of default definition, risk characteristic distribution, and lending standards.

The creation of homogeneous risk classes is a rigorous process that involves in-depth analysis of risk factors, consistent exposure assignment, continuous monitoring, and the use of representative data. At this stage, no specific requirements regarding the grouping approach are outlined, which provides a wide margin of flexibility in the creation of classes that must align with the regulator's expectations.

1.2.2 Focus on the Innovative Approach Involving Dynamic Programming

The document by Mohammad G. Mostafa Khan, Niraj Nand, and Nesar Ahmad (Khan, Nand, and Ahmad 2008) focuses on determining optimal stratum boundaries (OSB) using a dynamic programming approach, particularly for variables following a triangular or standard normal distribution. The objective is to minimize the variance of the population estimator by finding the best boundaries to create homogeneous strata. The dynamic programming approach for determining OSB is based on the following points:

- The problem of determining OSB is reformulated as a problem of determining optimal stratum widths (OSW), which can be solved using mathematical programming.

- Formulation of the mathematical programming problem (MPP): The MPP consists of minimizing the variance of the estimated population parameter, using Neyman allocation, under the constraint that the sum of the stratum widths equals the total range of the distribution.

- Variable studied: The variable studied (X) is considered as following a continuous distribution with a probability density function $f(x)$, which can be triangular or standard normal.

- Principle of dynamic programming: The MPP is treated as a multi-stage decision problem. At each stage, the value of the OSW, and thus the OSB for a stratum, is determined using a recurrence equation. The dynamic programming method applies Bellman's principle of optimality.

Key Mathematical Formulas and Notations:

- Let X be a variable studied with a probability density function $f(x)$ over the interval $[a,b]$.
- The strata are defined by the intervals $[x_{h-1}, x_h]$, where h is the index of the stratum, ranging from 1 to L.
- The width of stratum h is defined by $y_h = x_h - x_{h-1}$.
- The stratified mean is given by $\bar{x}_{st} = \sum W_h \bar{x}_h$, where W_h is the weight of stratum h, and $\bar{x}_h$ is the sample mean of stratum h.
- The variance of the population estimator is given by $V(\bar{x}_{st}) = \sum \frac{W_h^2 \sigma_h^2}{n_h}$, where σ_h^2 is the variance of stratum h and n_h is the sample size of stratum h.
- The optimal stratification problem consists of minimizing the variance $V(\bar{x}_{st})$ by finding the optimal boundaries x_h, given an allocation of the sample.
- In the case of Neyman allocation, the goal is to minimize the weighted sum of standard deviations of each stratum, i.e., $\sum W_h \sigma_h$.
- The objective function (ϕ) is defined as the variance function of stratum h, i.e., $\phi_h(x_{h-1}, x_h) = W_h \sigma_h$.
- The general minimization problem is formulated as follows:

$$\text{Minimize} \sum \phi_h(y_h), \text{ subject to} \sum y_h = d,$$

where d is the total range of the distribution.

The dynamic programming approach determines the optimal stratum boundaries by reformulating the problem as an MPP to minimize variance, solved recursively by determining the optimal stratum widths. The algorithm is applicable to continuous distributions such as the triangular and standard normal distributions, offering an efficient and flexible method for optimal stratification. As a result, this approach can inspire the creation of homogeneous risk classes.

1.3 Review of the definition of margin of conservatism

The margin of conservatism in credit risk modeling is a concept introduced to deal with uncertainties and potential errors in default probability estimates. It is designed to ensure that banks maintain a prudent approach to risk management and capital allocation (Branco 2011; Jongh et al. 2017). Basel II recognizes that estimates of default probability are subject to unpredictable errors, and requires banks to add a margin of conservatism to their estimates to avoid over-optimism (Jongh et al. 2017). However, the implementation of the MoC has caused some confusion among banks and regulators due to the various prudential requirements set out in Basel II (Jongh et al. 2017).

The EBA's guidelines on estimating the probability of default and losses in the event of default, require banks to put in place a framework for adjusting and correcting uncertainties linked to shortcomings in data, systems and methods (Liu 2018). This framework should be part of the rating and risk reporting process. In addition, various approaches have been proposed, including the use of upper confidence bounds for estimating the probability of default, while maintaining the order of credit quality indicated by the ratings (Tasche and Pluto 2004). The implementation of the MoC must take into account factors such as sample characteristics, calibration assumptions, business cycle effects and stress test results (Branco 2011).

1.3.1 Expectations regarding margins of conservatism

The MoC is not a measure of precision and does not attempt to indicate a range of possible values for the risk parameter. It is a quantity that is added to the best estimate of the risk parameter after the calibration phase, and after taking appropriate adjustments into account. These margins are designed to account for uncertainties and potential deficiencies in the data and methods used for estimating risk parameters, such as the probability of default and loss given default. Here are the key aspects related to the creation of margins of conservatism (EBA 2017; EBA 2016):

- Identification of deficiencies : Institutions must identify all deficiencies related to the estimation of risk parameters that could lead to bias in quantifying these parameters or increased uncertainty not fully covered by general estimation error.
- Categorization of deficiencies : These deficiencies are classified into three main categories:
  - Category A : MoC related to deficiencies in data and methods.
  - Category B : MoC related to changes in underwriting standards, risk appetite, recovery policies, etc.
  - Category C : MoC related to general estimation error.
- Appropriate adjustments : Institutions must apply adequate methodologies to correct the identified deficiencies and improve the risk parameter estimation.
- Quantification of MoC : MoC must reflect the additional uncertainty caused by the adjustments or deficiencies and must be quantified separately for each category.
- Final MoC : The final MoC is the sum of the MoC for each category, added to the best estimate of the risk parameter, without lowering the estimates.
- Monitoring and documentation : Institutions must regularly monitor MoCs and document the methods used for their application.
- Application of MoC : MoC should cover any deficiencies identified during the estimation process without biasing the risk quantification.
- Objectives and use : MoCs aim to limit unjustified variability in capital requirements while preserving the sensitivity of risk parameters to risk.

From this, it is ensured that risk estimations remain conservative and reflective of true risk levels, while also ensuring that capital reserves are sufficient to withstand any uncertainties or potential shortcomings.

1.3.2 The k-sigma Method for the Margin of Conservatism: Analytical and Bootstrap Approaches

There is no single formula for calculating the Margin of Conservatism (MoC), allowing financial institutions to adapt it to their needs. However, several methods exist to estimate this margin, particularly for the Probability of Default (PD). Among them, the k-sigma method is widely used (Gennaio 2019). This method adjusts the risk parameter by multiplying the estimator's standard deviation by a calibration factor k:

$$\mathrm{MoC} = k \times \sigma$$

The standard deviation σ reflects the statistical dispersion of the estimator and can be calculated using different approaches. The first relies on the volatility of the calibration sample, where σ is determined from the standard deviation formula of a binomial distribution:

$$\sigma = \sqrt{\frac{p(1-p)}{n}}$$

where p represents the central tendency of the calibration (Long Run Average Default Rate, LRADR), and n is the sample size.

Another approach evaluates the intra-variance of the estimates. This method decomposes the total variance into intra and inter components. The intra-variance is calculated as follows:

$$s^2 = \frac{1}{N-1} \sum (x_i - \hat{x}_j)^2$$

where x_i denotes an observation, and $\hat{x}_j$ represents the estimated risk factor. For PD, applying the Brier score allows adjusting this variance by considering the gap between estimated default probabilities and observed default rates (Gennaio 2019). The standard deviation is then computed as:

$$\sigma = \sqrt{\frac{\sum (PD_j - DR_j)^2 \cdot N_j}{N-1}}$$

where PD_j represents the estimated default probability for rating class j, DR_j is the observed default rate for the same class, and N_j is the sample size in that class. This approach integrates the uncertainty of estimates, refining the volatility measure of the estimator. It thus serves as a basis for determining a more robust Margin of Conservatism.

The value of the k factor can be obtained through different approaches. An analytical solution involves applying mean and variance formulas to define the total loss distribution, thereby adjusting k according to the 99.9% quantile. Alternatively, a Monte Carlo simulation can be conducted. This numerical approach generates random values for risk parameters, providing a more empirical estimate of k. Applying the MoC must consider several key principles. It should be applied at the calibration segment level rather than at the individual rating level. Additionally, the estimation error should be inversely proportional to the sample size, and special attention should be given to avoiding double counting of risks.

In addition to the analytical k-sigma method, an alternative approach relies on bootstrap, a non-parametric method that estimates uncertainty without prior assumptions about data distribution. This technique generates multiple samples by resampling with replacement. PD is recalculated at each iteration, allowing the estimator's distribution to be estimated and its standard deviation to be derived. The MoC calculation via bootstrap follows a rigorous procedure: observations are randomly sorted, and the first sub-samples are selected. The mean PD is computed for each sample, and the standard deviation between means is estimated. Only the right-hand side of the distribution (corresponding to values higher than the overall mean) is retained for semi-variance calculation.

The MoC application via this approach can follow an additive or multiplicative definition:

$$\text{MoC}_{add} = \max\left[x \times \left(\frac{\bar{x} + \sigma_{right}}{\bar{x}} - 1\right), 0.01\%\right]$$

$$\text{MoC}_{mult} = \frac{\bar{x} + \sigma_{right}}{\bar{x}}$$

Thus, we obtain:

$$\text{PD}_{MoC} = \text{PD} + \text{MoC}_{add} = \text{PD} \times \text{MoC}_{mult}$$

The bootstrap approach has several advantages. It does not require any assumption about data distribution and directly estimates the estimator's uncertainty. It is particularly useful when data do not follow a known distribution or when the sample size is limited. However, it assumes the independence of observations, which may require adjustments in the presence of correlations. Thus, the MoC calculation for PD can be performed using various methods, with the analytical k-sigma method offering a simple and systematic approach, while bootstrap provides a more flexible and empirical uncertainty estimation.

2 Methodology

This section outlines the methodological approach adopted in this work, which is divided into two main steps.

First, the construction of risk classes (HRCs) is addressed based on the provided scores. Four approaches were implemented to achieve optimal HRCs in terms of risk homogeneity and differentiation. Once the optimal HRCs are selected, their stability in terms of volumetry and default rate is evaluated.

Second, margin of conservatisms are estimated to account for uncertainties associated with the model. These margins are added to the default rates obtained for each HRC to produce adjusted default rates, which form the basis for the final HRCs. This step focuses on the methodologies employed to estimate and apply these margins effectively.

2.1 Approach to the construction of Homogeneous Risk Classes

The construction of Homogeneous Risk Classes (HRC) consists of transforming a continuous risk score variable into an ordered discrete variable, where each category represents a distinct risk level. The goal is to facilitate the analysis and interpretation of risk while ensuring that the classes reflect homogeneous and distinct levels in terms of default rates.

Since the default variable is the target in a default risk scoring project, the construction of HRC becomes a supervised learning problem. The discretization aims to:

- Identify optimal thresholds for defining the classes,
- Ensure that each class is distinct in terms of risk,
- Guarantee homogeneity within each class.

In each of the four approaches to be presented, once the HRCs are obtained, a minimum of 1% of the population in each HRC and a 30% relative difference between HRCs must be ensured. If these two conditions are not met, adjustments should be made by merging HRCs until both conditions are satisfied.

2.1.1 CART Algorithm (Classification Tree)

The CART (Classification and Regression Trees) approach is based on a decision tree methodology where the target variable is the default status, and the explanatory variable is the risk score. This algorithm systematically segments the continuous risk score into disjoint intervals by minimizing an impurity criterion, such as entropy or the Gini index. By optimizing these splits, CART identifies thresholds that effectively separate classes based on their default rates. Thus, many metrics are used for class discrimination.

Impurity reduction

The **impurity reduction** of a node t, denoted as $\Delta i(t)$, is defined as the difference between the impurity of the node before the split and the weighted sum of the impurities of the child nodes after the split. Mathematically, it can be expressed as:

$$\Delta i(t) = i(t) - (p_L \cdot i(t_L) + p_R \cdot i(t_R))$$

where:

- $i(t)$ is the impurity of node t,

- $i(t_L)$ and $i(t_R)$ are the impurities of the left and right child nodes,
- p_L and p_R are the proportions of elements in the left and right child nodes relative to the parent node.

This impurity reduction is a key criterion in decision tree construction like CART, where the goal is to maximize $\Delta i(t)$ at each split to obtain the purest possible nodes. The impurity criterion $i(t)$ commonly used for CART classification tree are Gini diversity index and Shannon entropy.

Gini criterion

The **Gini diversity index**, or **Gini index**, of a discrete random variable Y taking values in $\{C_1, \ldots, C_M\}$ is defined as:

$$IG(Y) = \sum_{l=1}^{M} \sum_{\substack{k=1 \\ k \neq l}}^{M} p_l p_k$$

where p_l represents the probability of Y taking the value C_l.

The Gini index is widely used in decision tree algorithms as a measure of node impurity, with lower values indicating purer distributions.

Shannon criterion

The **Shannon entropy** of a discrete random variable Y taking values in $\{C_1, \ldots, C_M\}$ is defined as:

$$H(Y) = -\sum_{l=1}^{M} p_l \log p_l$$

where p_l is the probability of Y taking the value C_l.

Given a sample $\{y_i\}_{i=1}^{n}$, the empirical estimation of entropy is given by:

$$\hat{H}(Y) = -\sum_{l=1}^{M} \hat{p}_l \log \hat{p}_l$$

where $\hat{p}_l$ is the empirical probability of C_l in the sample.

Entropy is commonly used in information theory and decision trees to measure uncertainty or impurity in a distribution.

Among its key advantages, the CART method offers flexibility in capturing non-linear relationships between the risk score and default rates, allowing for more nuanced decision boundaries. Additionally, it does not impose strong assumptions about the distribution of the risk score, which enhances its adaptability to diverse datasets. However, the approach is not without limitations. It is highly sensitive to hyperparameters such as the maximum tree depth and the minimum number of observations per class. Moreover, CART models are prone to overfitting, particularly in the absence of proper regularization techniques.

2.1.2 Weight of Evidence (WOE)

The Weight of Evidence (WOE) is a weighted logarithmic transformation that measures the heterogeneity between two groups (default vs. non-default) and is defined as follows:

$$WOE = \log\left(\frac{P(\text{non-default} \mid \text{class})}{P(\text{default} \mid \text{class})}\right)$$

WOE plays a crucial role in discretization, as it assesses the separation between classes in terms of default rates and helps identify optimal thresholds by maximizing the differences between groups. This approach offers several advantages, including excellent interpretability and the direct assessment of class homogeneity. However, it also has limitations, as it requires a good balance between default and non-default populations and is sensitive to highly imbalanced distributions.

In this study, a predefined number of classes is established (10 in this case, though this choice is open to discussion). For each class, the WOE is calculated using the formula above. Once the WOE is determined for all classes, HRCs with similar or closely aligned WOE values are grouped together. The process is then refined iteratively, evaluating the two conditions outlined earlier until both are satisfied.

In the context of risk management, as demonstrated in this study and other related works, the CART algorithm ultimately defines a set of terminal nodes, or leaves, which are interpreted as HRCs. These HRCs provide a structured and interpretable segmentation of the risk profile.

2.1.3 Quantile Approach

In credit risk modeling, the quantile approach is a widely used method for constructing Credit Risk Classes based on predicted scores. This technique segments the distribution of predicted credit scores into distinct quantiles, with each quantile representing a HRC. By categorizing borrowers into different risk levels, the method facilitates more precise risk assessment and management.

The quantile approach is well-documented in the literature for its effectiveness in credit risk assessment. For example, the Basel Committee on Banking Supervision emphasizes the importance of accurate risk differentiation in credit risk modeling Bank for International Settlements. Additionally, Nehrebecka (Nehrebecka 2019) highlights the use of scorecards to differentiate between good and bad firms by estimating the Probability of Default (PD) over a specified period Bank for International Settlements. These studies underscore the utility of quantile-based segmentation in enhancing the predictive accuracy of credit scoring models.

This approach has several advantages. It is simple and easy to implement, requiring minimal computational effort while being intuitive and transparent due to its reliance on straightforward statistical partitioning. Furthermore, it provides an objective classification method by using statistical thresholds (percentiles), which minimizes subjectivity. The method is particularly useful when equal representation across HRCs is desirable.

However, the quantile approach also presents limitations. It may lead to unequal group sizes, making HRCs potentially heterogeneous. Since it relies solely on statistical partitioning, it ignores domain-specific insights and expert knowledge that could enhance the relevance of thresholds. Additionally, it is less granular at the distribution tails, where risks are typically most critical, and it is sensitive to extreme values or outliers, which can distort the quantile division.

As with the Weight of Evidence (WOE) method, the quantile approach requires the selection of a fixed number of classes, typically 10 or 20. Based on the resulting HRCs, the two conditions previously mentioned must be verified. If necessary, HRCs with similar default rates are merged, and this process is iterated until both conditions are satisfied.

2.1.4 Dynamic Programming

This approach is based on a parametric method where only the characteristics of the continuous variable are considered. The objective is to minimize an error criterion, such as intra-class variance, using dynamic optimization. The method relies on certain assumptions, including that the variable follows a specific distribution, often normality, and that the target variable is not directly considered. Its advantages include simplicity, speed, and a well-defined algorithmic structure. However, it has limitations, such as the potential invalidity of the normality assumption and the lack of direct consideration of the target variable. In this study, the Biased Random-Key Genetic Algorithm (BRKGA) will be used. A brief overview of this approach will be given before presenting the results.

The Biased Random-Key Genetic Algorithm (BRKGA) is a metaheuristic[1] optimization method inspired by the biological evolution of species. It is designed to solve complex optimization problems by evolving a population of solutions through selection, crossover, and mutation. BRKGA starts with an initial population of candidate solutions, each represented by a vector of random numbers between 0 and 1, known as keys. These keys are then decoded to obtain feasible solutions to the optimization problem. Evolutionary Process is done in the following way :

- **Selection:** The best solutions from the current population are selected to survive to the next generation.
- **Crossover:** New solutions are generated by combining selected solutions with others from the population using a uniform crossover mechanism. A probability parameter controls the bias towards inheriting genes from elite parents.
- **Mutation:** A small portion of the population is replaced with randomly generated solutions to maintain diversity.

Once the HRC have been defined, as with previous approaches, it is crucial to assess their stability based on two key criteria. First, the volumetric criterion ensures that each class represents at least 1% of the total population. Second, the default rate criterion requires that the relative differences between two adjacent classes be significant (at least 30%) and that default rates follow a monotonic trend across the HRC. Additionally, these criteria must remain satisfied over time.

[1] A metaheuristic is a high-level problem-independent algorithmic framework that provides a set of guidelines or strategies to develop heuristic optimization algorithms. It is typically used to find approximate solutions to complex optimization problems.

2.2 Approaches to assess the homogeneity and the heterogeneity of risk classes

Before proceeding to the estimation of Margin of conservatism(MoC), it is essential to ensure that the different risk classes constructed using the different approaches from the previous section are homogeneous within and heterogeneous between one another. The objective is to minimize risk estimation errors for an individual assigned to a given class, thereby ensuring that the classes are clearly distinguishable in terms of default risk (default rates) and that they contain individuals with highly similar risk profiles.

To achieve this, various statistical tests can be used. This section aims to present the theoretical framework and the underlying hypotheses supporting these tests.

2.2.1 Heterogeneity of risk classes

The heterogeneity of risk classes is based on the principle that if they are truly distinct with respect to a given variable, the distribution of this variable should differ across classes. The choice of statistical tests depends on the nature of the variable, with the homogeneous risk classes serving as the reference.

Chi-square test

The Chi-square test is used when the risk driver considered for assessing heterogeneity is a qualitative variable. This test determines whether there is an association between the risk driver and the reference variable (HRC). If a significant association is found, the HRCs will be considered heterogeneous with respect to this risk driver.

The Chi-Square test assesses the independence between qualitative variables by comparing observed and expected frequencies.

Hypotheses

- H_0: The two variables are independent.
- H_1: The two variables are dependent.

The statistic of this test and its p-value are given as follow.

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad p\text{-value} = P(\chi^2 > \chi^2_{\text{obs}})$$

where:

- O_{ij}: Observed frequency in cell (i, j) of the contingency table.
- E_{ij}: Expected frequency in cell (i, j) under the null hypothesis, computed as:

$$E_{ij} = \frac{\sum_j O_{ij} \sum_i O_{ij}}{N}$$

  where N is the total sample size.
- χ^2_{obs}: The observed Chi-square value computed from the sample data.
- $(r-1)(c-1)$: Degrees of freedom, where r is the number of rows and c is the number of columns in the contingency table.

- $P(\chi^2 > \chi^2_{\text{obs}})$: The probability of obtaining a test statistic at least as extreme as the observed value, under the null hypothesis.

If the p-value is less than α, which can be 0.01 or 0.05, then the null hypothesis is rejected, and the HRCs are considered heterogeneous with respect to the risk driver involved.
This test has some advantages, such as being simple to compute and widely used, but it requires a sufficiently large sample size and expected frequencies $E_{ij} > 5$.

Welch test

The Welch test is a statistical test used when one of the two variables being studied is quantitative and the other is qualitative. In the context of this study, this test will be used when the risk driver is a quantitative variable. Since the risk drivers, for scoring purposes, are discretized and thus qualitative in nature, the only variable in this case will be the score provided for this study.

This test is a variation of the Student's t-test and is used when the assumption of equal variances between the two populations is not met. Therefore, it will be applied to all pairs of homogeneous risk classes.

Hypotheses

- H_0: The two groups have equal means.
- H_1: The means are different.

In oder to perform the test, we need to calculate the statistic of the test and its p-value. For this,the following formula can be used.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

where:

- $\bar{x}_1$ and $\bar{x}_2$ are the sample means of groups 1 and 2,
- s_1^2 and s_2^2 are the sample variances of groups 1 and 2,
- n_1 and n_2 are the sample sizes of groups 1 and 2.

The degrees of freedom (df) associated with this test statistic are approximated using the Welch-Satterthwaite equation:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}},$$

where: $n_1 - 1$ and $n_2 - 1$ are the degrees of freedom for each sample variance.

The p-value is calculated using the same approach as the Chi-square test, replacing the statistics with the t-statistic. If the p-value is smaller than the predetermined threshold α , the null hypothesis is rejected, indicating that the means of the two groups are different. This conclusion implies that the Homogeneous Risk Classes (HRCs) are heterogeneous with respect to the risk driver involved.

This test is more robust than t-test since it handles inequal variances but is limited by the normality it assumes in both groups.

Kolmogorov-Smirnov test

Unlike the previous test, which assumes the normality of the quantitative variable within groups, the Kolmogorov-Smirnov (KS) test makes no assumptions about the distribution of the quantitative variable. It is a non-parametric test used to compare the empirical distributions of a quantitative variable between two groups. In the context of analyzing heterogeneity, the KS test is employed to assess whether the distributions of scores differ between pairs of Homogeneous Risk Classes (HRCs).

Hypotheses

- H_0: The two samples follow the same distribution.
- H_1: The distributions differ.

The statistic of this test is calculated using the following formula.

$$D = \sup_x |F_1(x) - F_2(x)|$$

where $F_1(x)$ and $F_2(x)$ are the empirical cumulative distribution functions (ECDFs).

A p-value lower than the threshold indicates that the distributions differ, meaning the pairs of groups are distinct, and thus the Homogeneous Risk Classes (HRCs) are heterogeneous. This test is non-parametric and can detect any type of distribution. Its limitation is that it is sensitive to sample size; larger samples can lead to the detection of trivial differences, while smaller samples may not have sufficient power to identify meaningful differences.

Analysis of Variance (ANOVA)

ANOVA is a statistical method used to study the relationship between a quantitative variable and a qualitative variable. It aims to determine whether the means of the quantitative variable are equal across the different groups defined by the categories of the qualitative variable. Essentially, ANOVA is used to assess if the quantitative variable influences the qualitative one. If so, the average behavior or mean of the quantitative variable will differ in at least one of the groups.

Hypotheses

- H_0: All group means are equal.
- H_1: At least one group mean is different.

$$F = \frac{\text{variance}_{\text{inter}}}{\text{variance}_{\text{intra}}}$$

where:

- $\text{variance}_{\text{inter}}$ (Mean Square Between): Measures the variability due to differences between the group means.

$$\text{variance}_{\text{inter}} = \frac{\sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2}{k-1}$$

with:

- k: Total number of groups.
- n_i: Number of observations in group i.

- $\bar{x}_i$: Mean of observations in group i.
- $\bar{x}$: Overall mean of all observations.
- $\text{variance}_{\text{intra}}$ (Mean Square Within): Measures the variability within each group.

$$\text{variance}_{\text{intra}} = \frac{\sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2}{N - k}$$

with:

- x_{ij}: Value of observation j in group i.
- N: Total number of observations in the study ($N = \sum_{i=1}^{k} n_i$).

The degrees of freedom associated with each variance are:

- $\text{variance}_{\text{inter}}$: $k - 1$
- For $\text{variance}_{\text{intra}}$: $N - k$

The p-value is obtained from the F-distribution with $(k-1)$ and $(N-k)$ degrees of freedom. A small p-value (typically less than 0.05 or 0.01) suggests that the group means are significantly different and then HRCs are heteregenous.

2.2.2 Homogeneity of risk classes

For homogeneity, nearly the same tests are used except for the Z-test. Indeed, for homogeneity, we now focus within the HRCs, and two complementary approaches are employed. The first approach involves using the scores as the reference variable, and for each risk driver (a qualitative variable), ANOVA is performed within each risk class. Other tests, such as the Welch test and the Kolmogorov-Smirnov test, can also be applied. A class will be considered homogeneous with respect to a risk driver if the chosen test (ANOVA) leads to the acceptance of the null hypothesis.

The second approach uses the default variable as the reference variable. Within each HRC, for each risk driver, a Chi-square test is performed. For the score variable, the Kolmogorov-Smirnov test, Welch test, and ANOVA (with the default variable) are applied. An additional step in this approach is to complement it with proportion tests of default or default rates (Z-test) for each risk driver within each HRC.

Since the other tests have already been introduced in the homogeneity section, only the Z-test will be discussed in this section.

Z-test

The Proportion Comparison Test or Z-test is a statistical test used to compare the proportion of a qualitative variable (in this case, the default rate) between two groups defined by a qualitative variable. There are two ways to implement this test.

The first approach considers the risk drivers as binary categories (0-1). In this case, for each HRC, the default rate is compared between the two groups defined by these two categories.

The second approach treats the risk driver as a non-dichotomized variable. In this case, the default rate is compared between groups defined by the various categories of the variable, taking the categories in pairs. For example, if a variable has n categories, the test will be implemented $\frac{n(n-1)}{2}$ times.

Hypotheses

- H_0: The proportions of the two groups are equal
- H_1: The proportions of the two groups are unequal

For the z-statistic, the formula below is used.

$$Z = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Where: - $\bar{x_1}$ and $\bar{x_2}$ are the sample means of the two groups. - σ_1 and σ_2 are the population standard deviations of the two groups (known). - n_1 and n_2 are the sample sizes of the two groups.

If the p-value is greater than α, the null hypothesis is accepted and concluded that there is no significant difference. The involved HRC is then homogeneous.

This test is simple to implement when the standard deviation is known. This last one represents a limit since, in many practical situations, this standard deviation is not known.

2.3 Implementation of margins of conservatism

The estimation of the Probability of Default (PD) is a crucial step in credit risk management. However, due to statistical uncertainties and data limitations, it is necessary to apply a Margin of Conservatism (MoC) to ensure a prudent estimation that meets regulatory requirements. Different approaches exist to quantify this margin, each with its own advantages and limitations depending on the context and available data. We present three main methods: the parametric approach using confidence intervals, bootstrapping with resampling, and the Bayesian approach.

2.4 Parametric Approach Using Confidence Intervals

A first method to define the MoC is to use confidence intervals constructed around the PD estimator. Let $\hat{PD}$ be the point estimator of the probability of default for a given risk class. The $(1-\alpha)$ confidence interval is defined as:

$$\left[\hat{PD} - z_{\alpha/2} \cdot \sigma_{\hat{PD}}, \hat{PD} + z_{\alpha/2} \cdot \sigma_{\hat{PD}}\right],$$

where $z_{\alpha/2}$ is the quantile of the standard normal distribution and $\sigma_{\hat{PD}}$ is the standard deviation of the estimator. The margin of conservatism can then be defined as:

$$MoC = z_{\alpha/2} \cdot \sigma_{\hat{PD}}.$$

2.5 Bootstrapping with Resampling

An alternative approach is to use bootstrapping to estimate the uncertainty surrounding $\hat{PD}$. This method involves generating B samples obtained through resampling with replacement, and recalculating the PD for each sample b:

$$\hat{PD}^{(b)}, \quad b = 1, \ldots, B.$$

From this empirical distribution, the margin of conservatism can be defined as the $(1-\alpha/2)$ quantile of this distribution:

$$MoC = q_{1-\alpha/2} - \hat{PD}.$$

2.6 Bayesian Approach

The Bayesian approach allows incorporating prior knowledge about the probability of default by modeling it as a random variable. A more advanced Bayesian approach employs Markov Chain Monte Carlo (MCMC) techniques to estimate the probability of default. Starting from a regression model specification:

$$\hat{Y} = \alpha_1 \times \text{Class}_1 + \alpha_2 \times \text{Class}_2 + \cdots + \alpha_p \times \text{Class}_p,$$

where the coefficients α_i represent the effects of different risk classes, we define weakly informative priors on α using a uniform distribution:

$$\alpha_i \sim \mathcal{U}(0,1), \quad \forall i \in \{1,\ldots,p\}.$$

The model is completed with a noise variance also following a uniform distribution:

$$\sigma^2 \sim \mathcal{U}(0,10).$$

The distribution of observations is then modeled as:

$$Y_i \sim \mathcal{N}\left(\sum_{j=1}^{p} \alpha_j \times \text{Mod}_{i,j}, \sigma^2\right), \quad \forall i \in \{1,\ldots,N\}.$$

2.7 Jeffreys Approach

The Jeffreys approach is a Bayesian method for constructing confidence intervals for a binomial proportion p. It uses the Jeffreys prior, which is a non-informative prior distribution, and derives the posterior distribution as a Beta distribution. The interval is calculated based on the quantiles of this posterior distribution.

For a given number of successes x out of n trials, the posterior distribution is:

$$p \mid x \sim \text{Beta}(x+0.5, n-x+0.5).$$

The $(1-\alpha)$ confidence interval for p is then given by:

$$[q_{\alpha/2}(x+0.5, n-x+0.5), q_{1-\alpha/2}(x+0.5, n-x+0.5)],$$

where $q_{\alpha/2}$ and $q_{1-\alpha/2}$ are the quantiles of the Beta distribution. Having the confidence interval, it is thus possible to determine the margin of conservatism by taking the difference between the upper quantile and the mean.

2.8 Student's Approach

The Student's approach for Margins of Conservatism (MoC) uses the Student's t-distribution to construct confidence intervals, particularly suitable for small samples where the population variance is unknown. Below are the key steps of this approach:

- **Estimation of the mean and standard deviation**:
  - Let $\hat{PD}$ be the estimated Probability of Default (sample mean).
  - Let s be the sample standard deviation.

- **Margin of Conservatism (MoC)**: The MoC is defined as half the width of the confidence interval:

$$MoC = t_{\alpha/2,\nu} \cdot \frac{s}{\sqrt{n}}.$$

  where:

  - $t_{\alpha/2,\nu}$ is the quantile of the Student's t-distribution for $\nu = n-1$ degrees of freedom.
  - n is the sample size.
  - s is the sample standard deviation.

This approach is particularly useful for small samples and provides a conservative estimate of the uncertainty around the PD.

2.9 Clopper-Pearson Approach

The Clopper-Pearson interval is a widely recognized method for constructing confidence intervals for a binomial parameter. Despite some caveats, it is often considered the gold standard (Reiczigel 2003). This approach relies on the inversion of exact binomial tests with equal-tailed acceptance regions. Below is a step-by-step explanation of the methodology:

- **Calculation of L_p**: L_p is the largest value such that:

$$B(n,p)(L_p) \leq \alpha/2.$$

  In other words, it is the maximum value for which the cumulative probability of observing L_p successes or fewer is less or equal than the half of the significance level.

- **Calculation of U_p**: U_p is the smallest value such that:

$$B(n,p)(U_p+1) > 1-\alpha/2.$$

  This means that it is the minimum value for which the cumulative probability of observing up to U_p+1 successes is greater than $1-\alpha/2$.

- **Construction of the confidence interval**: The $(1-\alpha)$ confidence interval includes all values of p such that:

$$L_p \leq \hat{p} \leq U_p,$$

  where $\hat{p}$ is the observed probability. This means that the interval contains all possible values of the binomial parameter p that would not be rejected by an exact binomial test with equal tails at the significance level α.

The Clopper-Pearson approach guarantees that the actual coverage probability of the confidence interval is at least equal to the nominal confidence level $(1-\alpha)$. However, this method is often criticized for being conservative, as it may produce wider intervals than necessary.

3 Data description

The data provided for the study comes from the company EY. It covers the period from 2007 to 2018 and contains individual-level variables, meaning variables specific to each individual. For example, financial variables like annual income (annual_inc) and other types of variables such as home ownership (home_ownership), grade, etc.

In total, the training dataset contains 1,695,479 observations and 171 columns. The test dataset contains 565,160 observations and 171 variables. It is worth noting that most of the variables are binary qualitative variables, as they have been dichotomized. Additionally, there are three time-related variables (`issue_d_date`, `earliest_cr_line_date`, `last_credit_pull_d`). A complete description of the variables can be found in the appendix of the document.

4 Explanatory Analysis

This section aims to present the descriptive statistics of the target variable "Target" corresponding to the default event as well as the score provided for the study. The goal of the final analysis is to assess the discriminatory power of the model and, therefore, the score.

4.1 Default Rate Analysis

The distribution of the "Default" variable, as shown in figure 1, highlights a rather imbalanced distribution with a relatively high default rate (13%). This elevated default rate is consistent with the portfolio under study, namely the retail portfolio. Over the years, the trend has been generally downward, with a default rate of over 25% in 2007 and a rate lower than 5% in 2018. Furthermore, a slight increase in the default rates is observed between 2013 and 2015, which could be explained by delayed defaults due to the effects of the 2008 financial crisis.

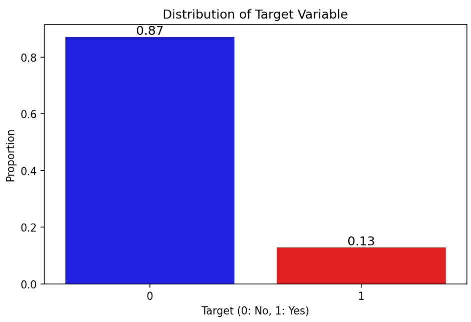

Figure 1: Default rate distribution

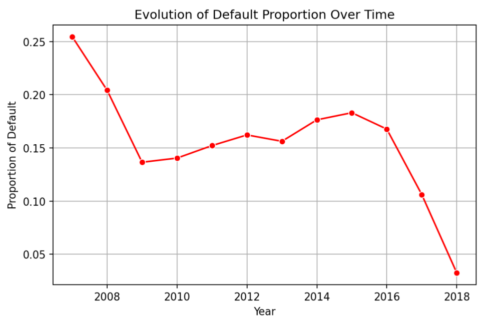

Figure 2: Default rate over time

4.2 Analysis of score

The score analysis reveals that the average score within the portfolio is 591, which is relatively decent. The observed scores range from a minimum of 399 to a maximum of 846. The relatively narrow score range suggests that the risk profiles of the observations may not differ significantly.

variable	Minimum	Median	Mean	Maximum
Score	399	583	590.9	846

Table 1: Statistic of Score

Figure 4 presents the density curves of scores for defaulting observations, non-defaulting observations, and the overall population. Since the curves for defaulting and non-defaulting observations are not well separated, it suggests that the scores do not effectively discriminate them.

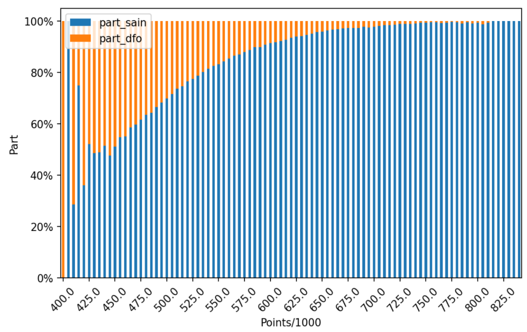

Figure 3: Score by default status

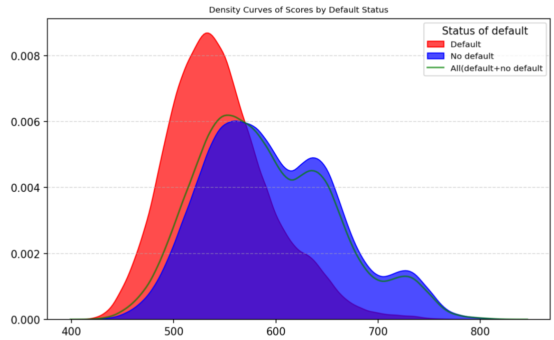

Figure 4: Density curve of score by default status

4.3 Analysis of the quality of the given model

The results on the discriminative capacity of the previously obtained scores highlight the importance to assess the quality of the given model. Therefore, it is crucial to conduct a detailed analysis of the model and monitor its performance over time. Such an evaluation would help make sense of the results in the construction of HRCs and the estimation of Margins of Conservatism(MoC).

The model provided is a logistic regression model. The dependent variable, called Target, takes the value 1 if the observation is in default and 0 otherwise.For the explanatory variables, the model uses a dichotomization approach, converting each category of each variable into a new binary variable (0-1). One category per variable is selected as the reference and is therefore excluded from the model.In total, 168 explanatory variables are used to estimate the logistic model provided for this study.

The model's quality was evaluated using several metrics, including the Area Under the ROC Curve (AUC)[2], the Gini Index[3], and the ratio of the Area Under the Precision-Recall Curve (AUC-PR)[4]. The results show that the model does not have good performance on both training and test samples since AUC is 0.74 on training and test samples, which gives a Gini Index of 0.48. This value is below 0.6, indicating the model's quality is considered weak, or at best, mediocre.

Additionally, by comparing the AUC-PR of 0.28 with a default rate of 0.13, we get a ratio of 2.15. This means the model is only about twice as good as a random model, which is still not sufficient for good prediction and classification.

[2]AUC measures the model's ability to distinguish between positive and negative classes. An AUC closer to 1 indicates better performance, while an AUC of 0.5 suggests the model is not better than the random model.

[3]The Gini Index is derived from the AUC and can be calculated by taking 2 times AUC and minus 1.It ranges from 0 to 1, where 1 represents perfect class separation and 0 means the model performs like random guessing.

[4]AUC-PR is a metric that helps evaluate the balance between precision and recall, especially when the classes are imbalanced.

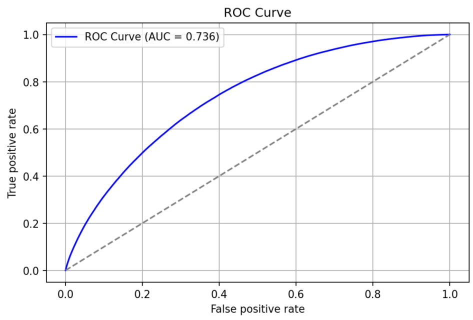

Figure 5: AUC train

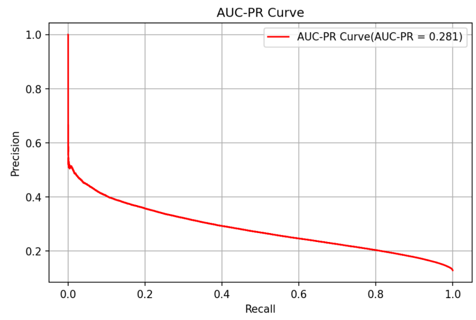

Figure 6: AUC-PR train

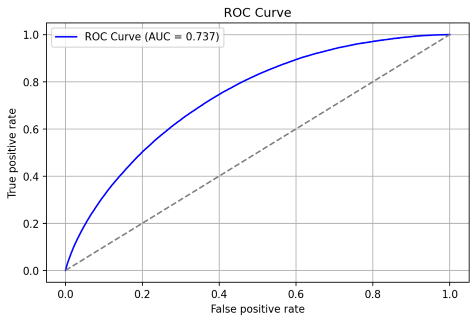

Figure 7: AUC test

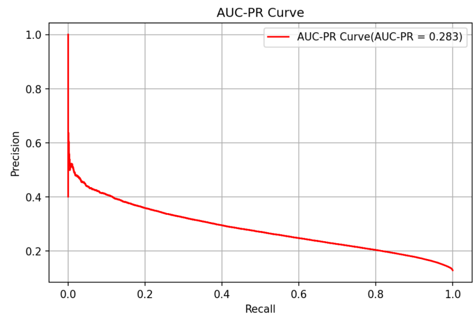

Figure 8: AUC PR test

The analysis over time shows that the model's performance has remained generally low. In fact, the Gini index has consistently stayed below 0.4 for all years.

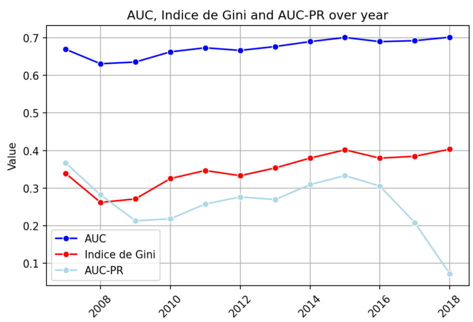

Figure 9: AUC, Gini index and AUC-PR on the train sample over time

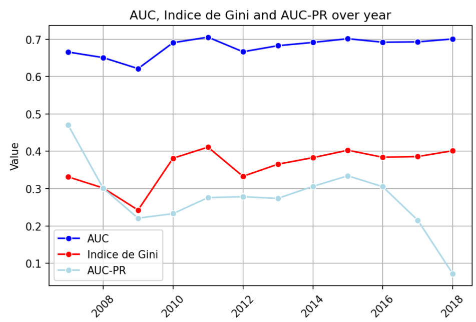

Figure 10: AUC, Gini index and AUC-PR on the test sample over time

The model is of poor quality, as shown by the analyses, and this could impact the construction of HRCs and the estimation of marges of conservatism.

With a clear view of the given model,the score and also the default distributions, we can now move forward with applying the various methodologies to construct our HRCs.

5 Homogeneous Risk Classification Obtained

By applying different approaches to creating Homogeneous Risk Classes (HRC), various results are obtained. This section aims to illustrate and compare them.

5.1 HRC obtained Using the CART Decision Tree Method

Here, we aim to construct HRCs using a supervised approach. For each obligor, we have the corresponding score and the target variable (good/bad borrower). A Classification and Regression Tree (CART) is trained using two impurity criteria: entropy and Gini index. To control the number of HRCs classes, the maximum depth of the tree is fixed. Additionally, a minimum of 5% of the obligor volume is required for each class to ensure sufficient representation. Initially, the model was applied with a maximum depth of 4, which theoretically allows for up to 16 HRCs classes. However, this resulted in 13 classes, and the relative deviation criterion for default rates across classes was not satisfied. Through manual adjustments, we aimed to reduce the number of classes to fewer than 8. Consequently, the models were retrained with a maximum depth of 3, limiting the number of HRC classes to a maximum of 8.

5.1.1 Gini criterion

When training a CART model using the Gini diversity index as the impurity criterion, we obtain 7 optimal HRC classes that satisfy both the volume and relative deviation criteria (see Table 2). The heterogeneity test (Welch and Wilcoxon tests with Bonferroni correction, p-value = 0) confirms significant differences across the HRC classes.

When computing the chi-square test between the default variable and the risk drivers for each HRC class sample, we found no statistically significant relationship for a certain number of risk drivers (table 26). This suggests that default is homogeneously distributed across these risk drivers in the considered sample. Consequently, the risk drivers that exhibit homogeneity across all HRC classes are the number of derogatory public records and the total number of credit lines currently in the borrower's credit file.

For a given HRC class and risk driver, the comparison of score distributions across the risk driver's categories using ANOVA (see table 23)reveals that only a few risk drivers remain homogeneous. Moreover, in some HRC classes (notably HRC 1 and HRC 2), no risk drivers exhibit homogeneity, highlighting potential limitations in their discriminatory power for these specific classes.

HRC Class	Thresholds	Default Rate (%)	Relative Deviation (%)	Gini index	Shannon entropy
1	> 647.5	2.39	0.00	0.047	0.113
2	607.5 - 6047.5	5.92	147.27	0.111	0.225
3	577.5 - 607.5	9.63	62.62	0.174	0.317
4	552.5 - 577.5	13.76	42.94	0.237	0.401
5	532.5 - 552.5	18.08	31.39	0.296	0.473
6	507.5 - 532.5	23.48	29.82	0.359	0.545
7	<=507.5	34.24	45.86	0.450	0.643

Table 2: HRC classes obtain with CART classification using Gini impurity criterion. Default Rate within HRC. Relative Default Rate Deviation across HRCs.

We can also observe that the stability of the empirical default rate and volume over the issue date is maintained from 2012 to 2017 (Figures 11 and 12). However, there appear to be migrations across HRC as illustrated in figure 40. Figure 11 highlights the volume within HRC and the corresponding observed default rate, showing that default risk significantly increases with HRC.

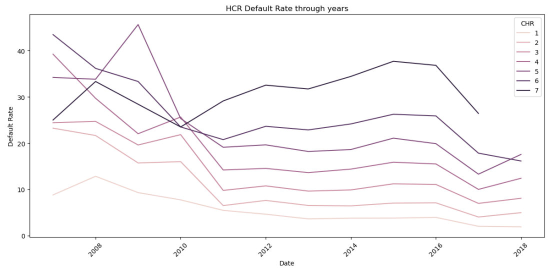

Figure 11: HRC Default rate through issue dates

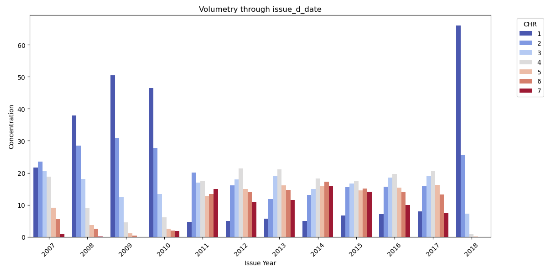

Figure 12: HRC volume through issue dates

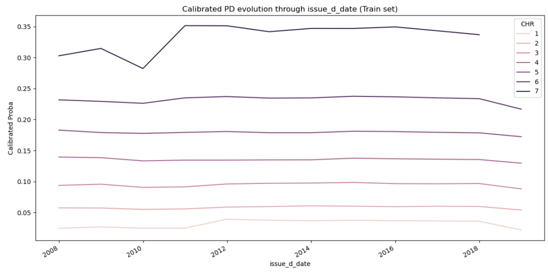

Figure 13: Calibrated PD through issue date.

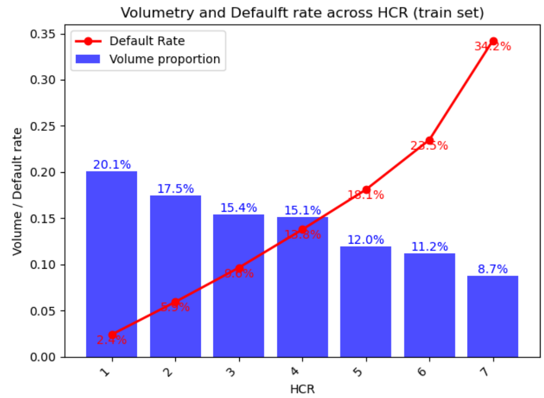

Figure 14: HRC volume and related default rate.

Given that stability fails, it appears that the score thresholds derived using the Gini index may not be optimal for validating all the required regulatory concerns regarding Probability of Default (PD). In the following section, we replace the Gini index criterion with Shannon entropy as the impurity criterion in the CART classification algorithm.

5.1.2 Shannon Entropy

Following the same steps as before, the optimization of the CART algorithm results in 8 optimal classes. As observed in Figure 16, stability is challenged during the periods 2008–2011 and 2017–2018. Meanwhile, the volume evolution between 2011 and 2017 appears more stable, despite occasional curve crossings over time. Additionally, the default rate curves of HRC 1 and 2 intersect at certain points (Figure 15), even though their relative deviation aligns with the expected threshold (Table 3). This suggests that these two classes should be merged. Furthermore, Figures 16 and 15 reveal a cutoff in HRCs after 2018, with some HRCs suddenly increasing while others decline. This is well illustrated by figure 41.

HRC Class	Thresholds	Default Rate (%)	Relative Deviation (%)	Gini index	Shannon entropy
1	> 687.5	1.31	0.00	0.0259	0.070
2	632.5 - 687.5	3.83	191.64	0.074	0.162
3	597.5 - 632.5	7.14	86.60	0.133	0.257
4	572.5 - 597.5	10.56	48.07	0.189	0.337
5	542.5 - 572.5	15.15	43.39	0.257	0.425
6	517.5 - 542.5	20.90	37.95	0.331	0.513
7	497.5 - 517.5	27.27	30.45	0.397	0.586
8	<= 497.5	36.94	35.45	0.466	0.659

Table 3: HRC class obtain with CART classification using Shannon entropy impurity criterion. Default Rate within HRC. Relative Default Rate Deviation across HRCs.

A comparison of default rates across HRC in Figure 18 and Figure 11 suggests that entropy-based HRC discriminates better than Gini-based HRC. This HRC meets the requirements for volume, relative default rate deviation (Table 3), and heterogeneity (p-value = 0 in Welch and Wilcoxon tests). However, score homogeneity given the target (good/bad borrowers) within HRC fails the Kolmogorov-Smirnov test.

Tables 24 and 27 present the results of the homogeneity tests for HRC across risk drivers and the score/target variable, respectively. Among the 22 initial risk drivers, HRC classes exhibit homogeneity with respect to 3 to 13 of them. The risk drivers commonly retained across HRC classes (with the exception of HRC 4) are: the number of derogatory public records and employment length in years (see Table 27).

In contrast, the ANOVA test applied to compare credit score distributions across risk driver classes within each HRC sample reveals fewer homogeneous risk drivers as shown in table 41. However, the chi-square test between the default variable and risk drivers yields satisfactory results in terms of homogeneity, indicating a consistent distribution of defaults across the identified risk drivers.

Given that we achieve better stability with Shannon entropy than with Gini, it might be interesting to proceed with the former. Before deciding whether to refine these results through further manual adjustments, we explore a weight-of-evidence-based approach in the following paragraph.

5.2 HRC obtained Through the Weight of Evidence Approach

As in the quantile approach, to ensure high granularity, the score variable was divided here into 50 quantiles before classes were created. Using the Chi-square statistic test, the algorithm uses this information to group the data. Upon reaching the desired number of 30 classes, the grouping loop ends.For these 30 classes, we calculate the WOE for each class. The graph below shows the differences in WOE between our 30 initial classes.

We set a threshold of 0.1 below which two classes that differ by a WOE of 0.1 are considered similar. Grouping is therefore based on this threshold to obtain 10 classes. The relative variation of 30% is achieved in our classes. Levene's test confirms a variation in variance between classes. Welch's statistic test and ANOVA, which are both significant (p-value = 0.00) and support the homogeneity within classes and heterogeneity between classes, are used in conjunction with this test. In the training and test datasets, this finding is noteworthy. Nonetheless, there are still certain issues with temporal stability and performance stability (Gini).

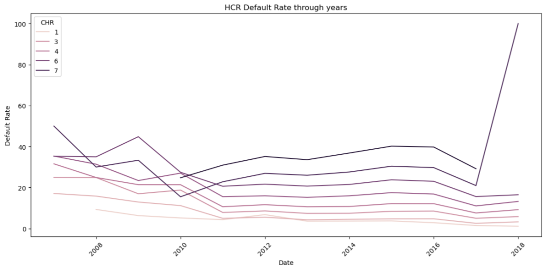

Figure 15: HRC Default rate through issue dates

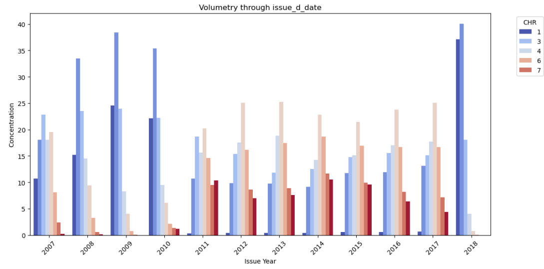

Figure 16: HRC volume through issue dates

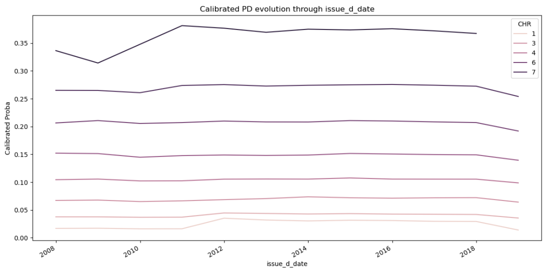

Figure 17: Calibrated PD through issue date.

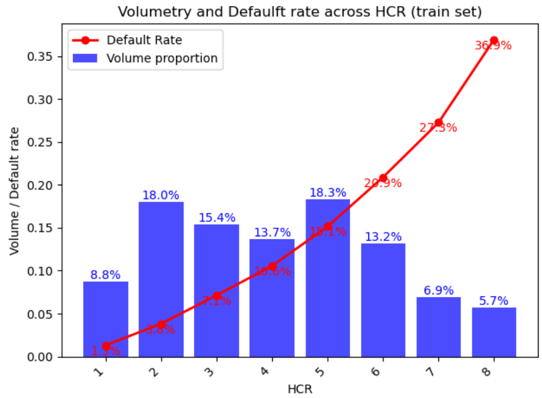

Figure 18: HRC volume and related default rate.

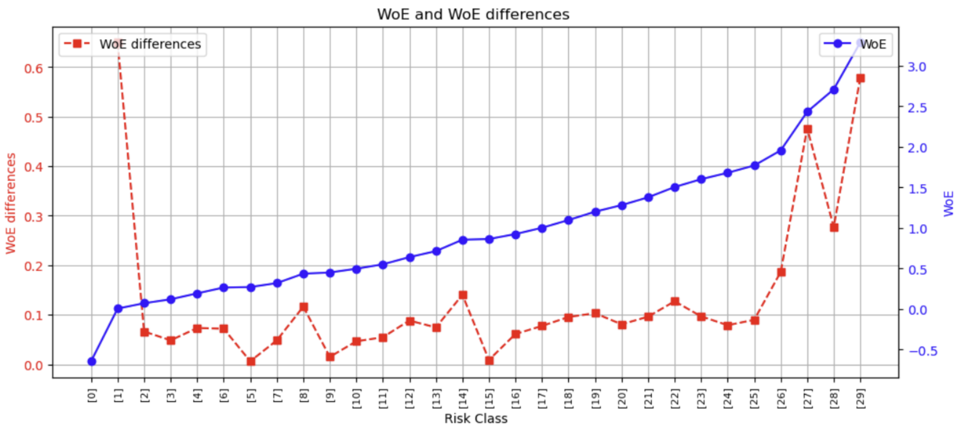

Figure 19: WOE and WOE differences for two sucessive classes

The Gini study shows very slight instability for the first 9 classes, but fairly pronounced instability for the last. The study of temporal stability in terms of volume and default rate shows crossover between classes, which points to limitations in either the approach, the PD modeling or the data structure.

5.3 HRC obtained via Quantile Discretization

The creation of classes was carried out by dividing the score variable into 40 quantiles to ensure high granularity. Based on this, the algorithm groups the data using the Chi-square test statistic. The grouping loop stops when the target number of classes is reached. In this case, the target number of classes was tested at 10, 12, 15, and 20. The results showed that 15 was an ideal target based on the representativity of individuals within the classes. However, these

Class	Global Default Rate (%)	Frequency (%)	Relative Variation (%)
1	0.55	1.93	
2	0.97	1.97	77.59
3	1.28	2.01	31.53
4	2.04	2.02	59.78
5	2.79	7.83	36.78
6	3.94	6.07	41.07
7	5.43	9.81	37.67
8	7.87	12.04	45.03
9	11.03	14.02	40.13
10	21.94	42.31	98.96

Table 4: Class summary.

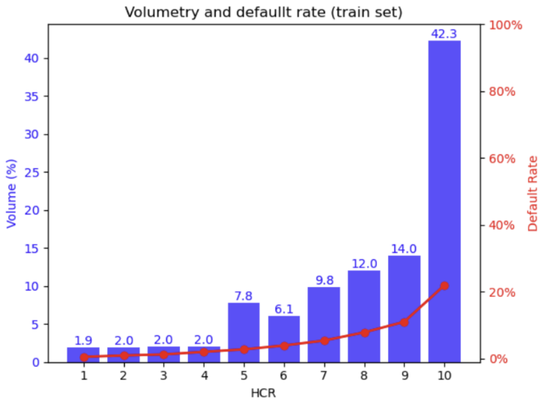

Figure 20: Distribution of Risk Classes and related default rate

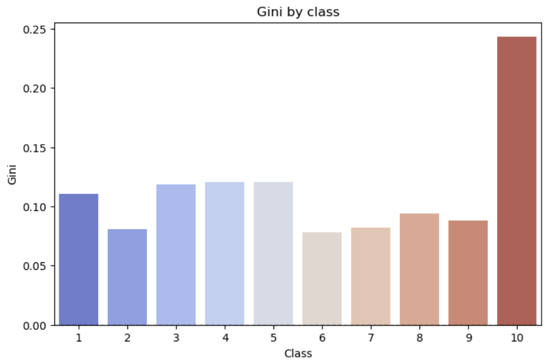

Figure 21: Observation of performance by class

15 classes did not fully meet several criteria defining a good risk classification. Specifically, the relative variation in the default rate between two successive classes was below 30% and, in some cases, as low as 5%. This proximity encouraged further class merging based on this criterion. As a result, the final classification consists of six classes, illustrated below.

Class	Global Default Rate (%)	Frequency (%)	Relative Variation (%)
1	0.61	2.48	0.00
2	1.07	2.45	77.43
3	1.74	2.49	61.79
4	2.70	7.31	55.58
5	5.07	19.86	87.52
6	17.70	65.41	248.96

Table 5: Class with Global Default Rate, Frequency, and Relative Variation in percentage.

This classification shows a difference in variance between classes, confirmed by Levene's test. This test is complemented by Welch's statistic test and ANOVA, both of which are also significant (p-value = 0.00), reinforcing the homogeneity within classes and heterogeneity between classes. This observation is notable in both the training and test datasets. However, some shortcomings remain regarding the stability of performance (Gini) and temporal stability.

It must be noted that there is a significant difference in Gini, particularly pronounced in the

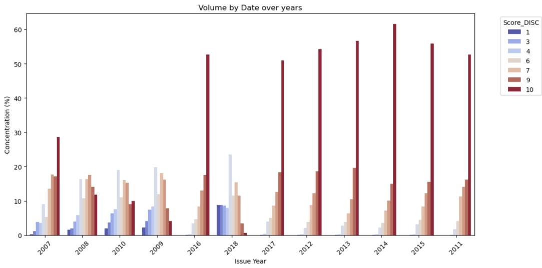

Figure 22: Evolution of volume over time

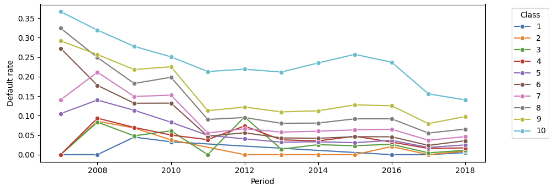

Figure 23: Evolution of the default rate over time

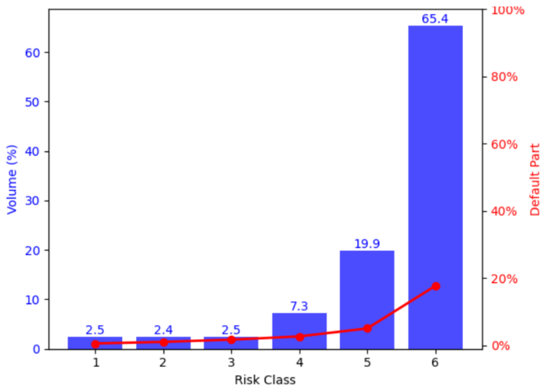

Figure 24: Distribution of Risk Classes and related default rate

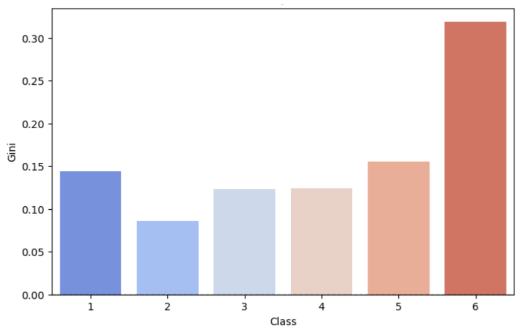

Figure 25: Observation of performance by class

highest-risk class. This difference indicates a lack of robustness in our classifications, which need to be improved using other methods. This instability is also observed when analyzing the default rate and volume trends over time.

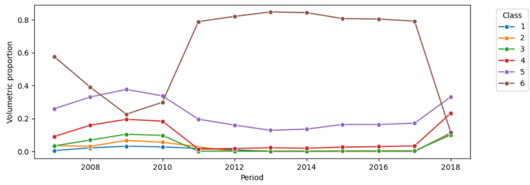

Figure 26: Evolution of volume over time

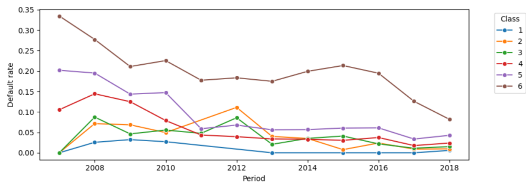

Figure 27: Evolution of the default rate over time

The results reveal instability in both the default rates and the class volumes over time. These findings suggest the need to further refine the creation of homogeneous risk classes using an approach such as dynamic programming.

5.4 HRC obtained Using Dynamic Programming : BRKGA

This method was applied with a predefined number of 10 classes. After obtaining the thresholds and HRCs, the different conditions for volume stability and default rate, as well as the relative differences in default rates, were evaluated. As all these conditions were not met, groupings based on similar observed default rates over time were made, resulting in five final HRCs.

Table 6 presents the different HRCs with their default rates, the proportions of observations they contain, and the relative differences in default rates. From this table, it can be observed that the relative differences in default rates exceed 30%, the default rates are monotonic across the HRCs, and all HRCs contain more than 1% of the observations.

HRC	Threshold	Default Rate (%)	Frequency (%)	Relative Variation (%)
1	[739;1000[	0.58	2.24	0.00
2	[659;739[	2.21	13.15	282.02
3	[617;659[	4.99	17.94	125.69
4	[576;617[	9.24	20.21	84.79
5	[399;576[	21.09	46.45	128.36

Table 6: HRC BRKGS

The evaluation of the stability over time of the default rates and volumes of these five HRCs is shown through the two graphs below. These graphs show that the HRCs are stable in default rates both on the training sample and the test sample (for which the default rate of HRC 4 surpassed that of HRC 5 in 2008). Volume stability is achieved for some HRCs, while others, such as HRC 5 and HRC 2, show instability over time.

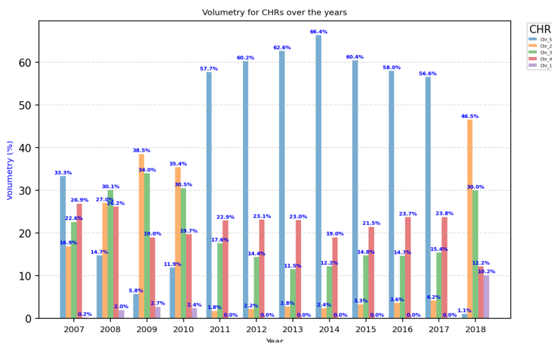

Figure 28: volume Stability train

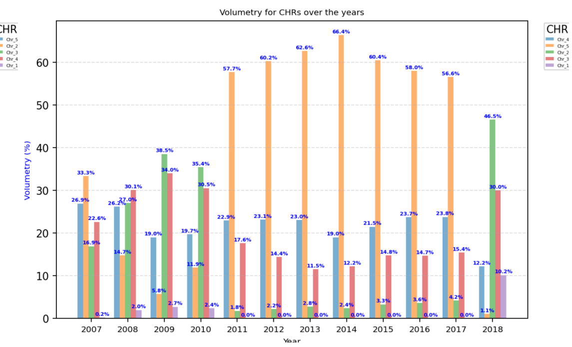

Figure 29: volume Stability test

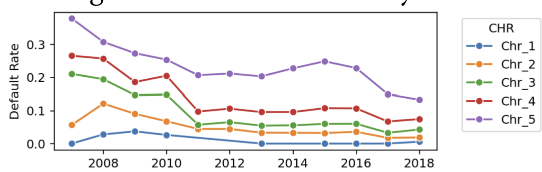

Figure 30: Default rate over time on train

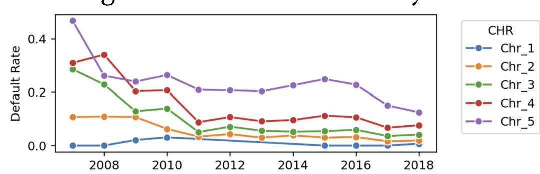

Figure 31: Default rate over time on test

To complement the stability analysis of default rates and volume, it was considered useful to further break down the analysis by HRC. Figures 32, 33, 34, 35, and 36 show that, overall, for the riskier HRCs (starting from HRC2), default rates exhibit a decreasing trend, similar to the global default rate trend observed over time during explanatory analysis. In HRCs 3, 4, and 5, the volume remains relatively stable. However, HRC1 shows an unusual volume pattern, with almost no volume between 2011 and 2017, which could be concerning but might also suggest a significant migration effect.

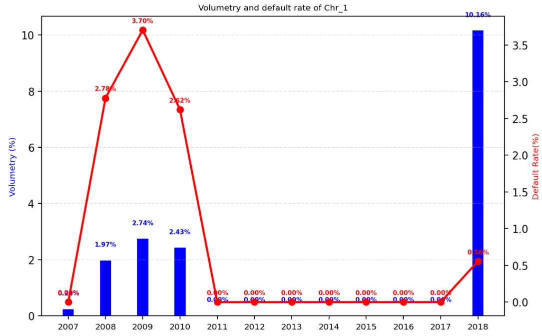

Figure 32: Stability HRC1

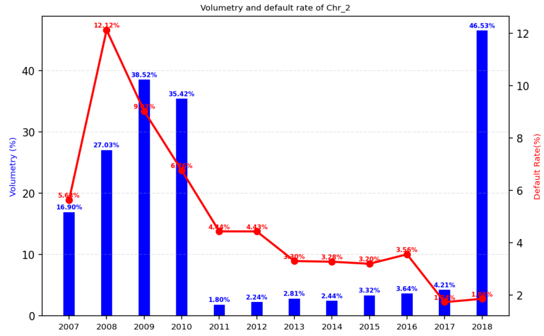

Figure 33: Stability HRC2

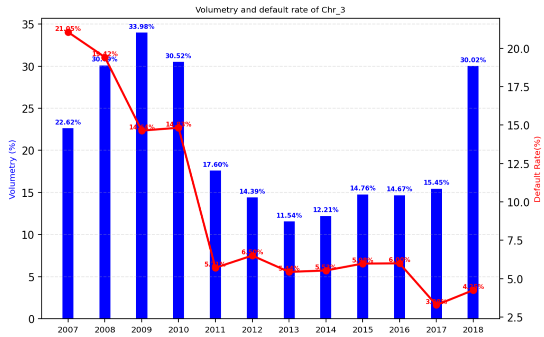

Figure 34: Stability HRC3

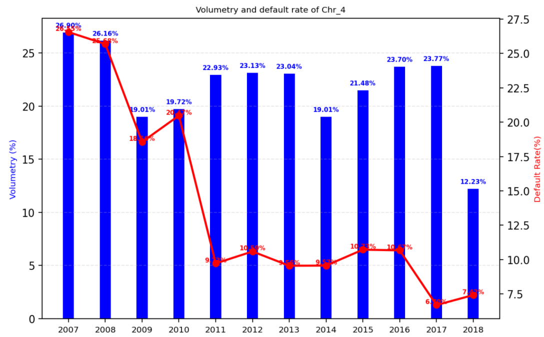

Figure 35: Stability HRC4

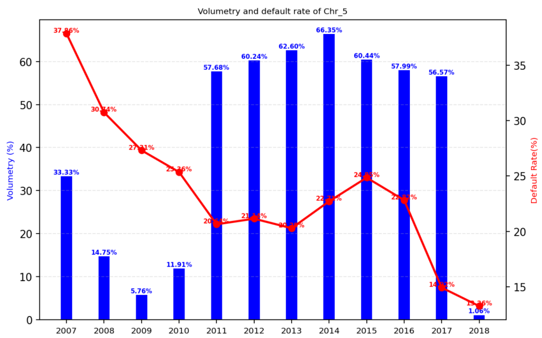

Figure 36: Stability HRC5

In addition to stability in volume and default rates, the HRCs must be homogeneous internally and heterogeneous between them. To achieve this, Chi-square, Kolmogorov-Smirnov, and Welch tests were performed on the training and test samples. Additionally, to verify if the HRCs remain well-calibrated to discriminate defaults over time, the PSI was calculated. The results of all these tests are recorded in the following figures.

The Welch test (table 6.4), with the score as the quantitative variable, is significant at the 5% (even at 1%) level on both samples(train and test), indicating that the score is distributed differently across the 5 HRCs.

The Chi-square tests (table 6.4) show that, overall, the HRCs are heterogeneous at the 5% level. In fact, only 4 variables (home ownership:OTHER, purpose vacation, addr state:AK, emp length:5-6) out of the 168 tested are homogeneous across the HRCs in the training sample (i.e., 5.88%), and 8 variables out of the 68 tested (home ownership:NONE, purpose vacation, addr state AK, addr state:DE, addr state:IA, addr state:NE, addr state:UT,addr state:WI) are homogeneous across the HRCs in the test sample(i.e, 11.6%).

Moreover, the Kolmogorov test is performed within each HRC and between the target and the score. Additionally, chi-square tests are conducted within each HRC to analyze the relationship between all risk drivers and the target variable. The results of these tests indicate that, among all variables, more than 43% share the same distribution between borrowers in default and those not in default within each HRC. This finding suggests that, within each HRC, borrowers are homogeneous with respect to most variables, confirming the homogeneity of HRCs. All tests are conducted with a 1% significance level.

HRC	#Homogen	#Heterogen	%Homogen
1	163	6	96.44
2	118	51	69.82
3	119	50	70.41
4	137	32	81.06
5	73	96	43.19

Table 7: Homogeneity of HRC Train

#Homogen	#Heterogen	%Homogen
164	5	97.04
138	31	81.65
140	29	82.84
150	19	88.75
100	69	59.17

Table 8: Homogeneity of HRC Test

#Homogen : Number of homogeneous variables.
#Heterogen : Number of heterogeneous variables.
%Homogen : Ratio of homogeneous variables.

The Z-tests have also been implemented, and the results are very similar to those presented in Table 8. The results for the Z-tests can be found in the appendix.

Furthermore, the PSI calculated on the training and test samples is 0.0002, which is strictly less than 0.1, indicating that the HRCs remain well-calibrated on these two samples.

To complement these analyses, Gini indices were calculated for the HRCs both overall and across the years. The Gini index is very low in the HRCs, and generally quite unstable over time.

HRC	Gini Index
1	0.13
2	0.23
3	0.14
4	0.12
5	0.26

Table 9: Gini Index - Train

HRC	Gini Index
1	0.09
2	0.24
3	0.16
4	0.13
5	0.26

Table 10: Gini Index - Test

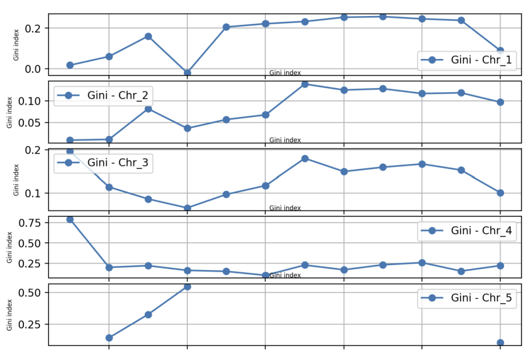

Figure 37: gini train over time

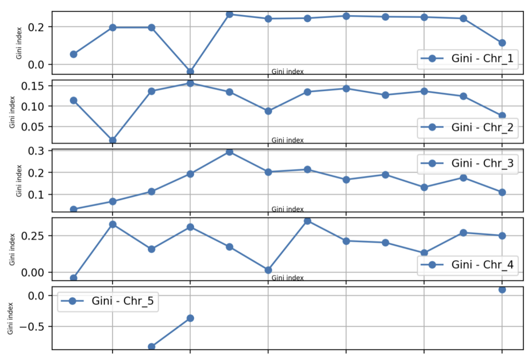

Figure 38: gini test over time

Figure 39 provides a representation of the model with the HRCs, their default rates, and their proportions.

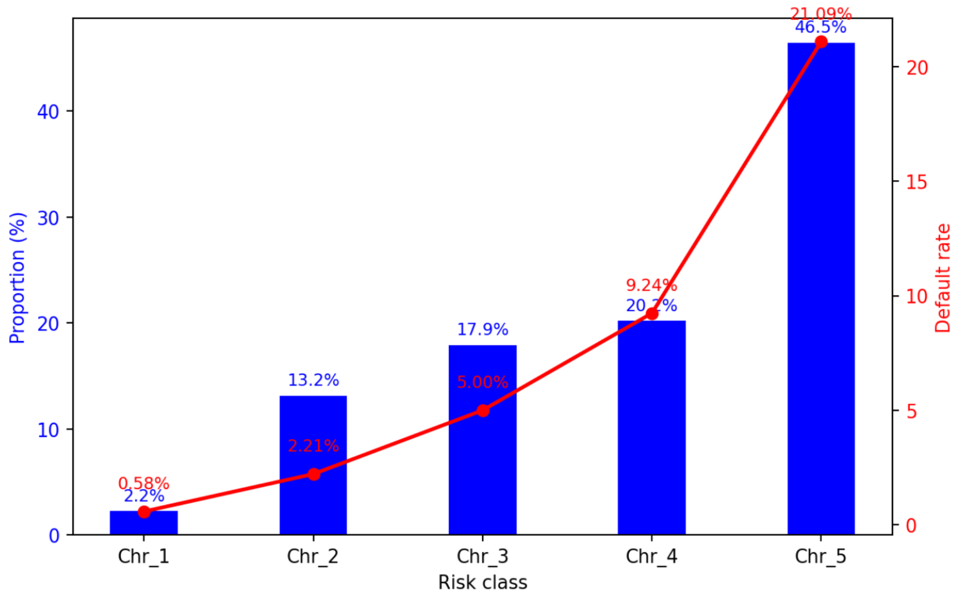

Figure 39: Model restitution

Different results are obtained depending on the approaches. However, due to the good separation observed through the CART approach with Gini and dynamic programming. The HRCs obtained with these two methods will be considered later in the context of defining the margins of conservatism.

6 MoC calibration

In the previous section, we discussed the construction of Homogeneous Risk Class segments using the CART method with the Gini index impurity criterion and the Dynamic Programming approach. Both methods yield highly accurate results, satisfying most regulatory constraints on the stability of calibrated Probability of Default (PD), concentration within HRC, heterogeneity across HRC, and homogeneity within HRC given certain risk drivers.

However, PD calibration can be affected by model risk, leading regulators to recommend the addition of a Margin of Conservatism (MoC) to the calibrated PD. According to Gennaio 2019, a framework for estimating MoC (Category C) emphasizes the following key points:

- The Basel framework fixes the confidence interval for PD parameters at 99.9% to protect against estimation errors. The MoC introduces an additional implicit increase in this already high confidence level.
- Application at Calibration Segment Level:

  MoC should be applied at the calibration segment level rather than the rating grade level. This is motivated by two main reasons:

  Capital Quantification: MoC affects capital quantification, which is determined at the calibration stage, the final layer in the parameter determination process.

  Avoiding Distortions: Applying MoC at the rating grade level could introduce distortions in risk measures, which may not be immediately identifiable.

In this section, we compare the MoC estimated using different approaches based on an aggregation on the HRC obtained. We consider several approcahes for MoC estimation: parametric approach, Boostrap and bayesian approach.

6.1 MoC on HRC obtained by the CART approach

The global Long Rate Average (LRA) PD is 12.86%, which represents the average PD across all HRC.

CART approach creates HRCs with balanced sizes, which is useful for ensuring that no single segment dominates the portfolio. The bootstrap MoC is 0.002199, and the parametric MoC is 0.001745 at the portfolio level (Table 11). These values indicate the overall uncertainty in PD calibration for the entire portfolio. The close agreement between the two methods suggests that both provide reliable estimates of the MoC.

HRC	Frequency (%)	LRA	MoC_bootstrap	MoC_param
1	8.7530	0.014887	0.000769	0.000896
2	18.0253	0.039071	0.000974	0.000999
3	15.3997	0.069828	0.001648	0.001421
4	13.7234	0.105294	0.001730	0.001813
5	18.3357	0.149716	0.002016	0.001824
6	13.1558	0.208715	0.002178	0.002452
7	12.6070	0.319088	0.006412	0.002873
Portfolio	100	0.128686	0.002199	0.001745

Table 11: MoC Calibration Results based on HRC obtained with CART approach.

The CART approach produces HRC segments with a clear risk gradient and stable MoC estimates. The bootstrap and parametric methods provide complementary measures of uncertainty, with the bootstrap method capturing slightly higher uncertainty for higher-risk segments. The results are consistent with regulatory requirements and provide valuable insights for capital allocation and model risk management

6.2 MoC on HRC obtained with dynamic programming

Table 12 presents the results of Margin of MoC estimation for HRC classes obtained using the Dynamic Programming approach. The bootstrap MoC (0.004791) is significantly higher than the parametric MoC (0.001438). The discrepancy between bootstrap and parametric MoC at the portfolio level suggests that the parametric method may not be suitable for this portfolio, especially given the presence of large, heterogeneous segments.

HRC	**Frequency (%)**	**LRA**	**MoC_bootstrap**	**MoC_param**
1	2.4837	0.008976	0.000170	0.001512
2	13.6261	0.024619	0.000854	0.001061
3	17.2269	0.050308	0.001254	0.001331
4	19.6250	0.090695	0.002030	0.001638
5	47.0384	0.209576	0.008622	0.001500
Portfolio	100	0.128624	0.004791	0.001438

Table 12: MoC Calibration Results based on HRC obtained with dynamic programming approach.

The Dynamic Programming approach creates unevenly distributed HRC segments, with one dominant segment (HRC 5) that exhibits high variability in PD estimates. The bootstrap MoC provides more conservative and reliable estimates of uncertainty, while the parametric MoC may underestimate risk, particularly for large, heterogeneous segments.

6.3 MoC obtained with Bayesian approach

The Bayesian approach involves the establishment of two Markov chains to execute the MCMC algorithm. However, the computational requirements lead to using only 1% of the entire dataset. Indeed, the approach requires so many resources that with a larger proportion, the program does not execute. Based on this, the approach was initiated, leading to the results presented below.

For HRCs obtained using the CART approach:

α	Mean (LRA)	MoC
α_1	0.0255138	0.0001950
α_2	0.0575668	0.0002004
α_3	0.0957498	0.0002152
α_4	0.1355552	0.0002254
α_5	0.1793849	0.0002534
α_6	0.2355564	0.0002665
α_7	0.3460295	0.0003025
Portfolio	0.1285547	0.0002281

Table 13: MoC Estimation for HRCs using the CART approach

For HRCs obtained using the dynamic programming approach:

α	Mean (LRA)	MoC
α_1	0.0146269	0.0004731
α_2	0.0239696	0.0002331
α_3	0.0502535	0.0002113
α_4	0.0929778	0.0001949
α_5	0.2137230	0.0001258
Portfolio	0.1295898	0.0001779

Table 14: MoC Estimation for HRCs using the dynamic programming approach

It is important to note that the MoC values are derived from Time-series Standard Error for two reasons:

- It accounts for temporal dependence between samples.
- It is generally larger than Naive Standard Error, as it adjusts the estimation by considering autocorrelation.

For regulatory compliance and capital allocation, the bootstrap method is preferred, but the underlying segmentation approach may need refinement to reduce model risk and improve the homogeneity of segments.

6.4 MoC obtained with Jeffreys, Student and Clopper-Pearson approaches

The Margin of Conservatism (MoC) calibration results based on HRC using three other statistical approaches: Jeffreys, Student, and Clopper-Pearson. The results are obtained using both the CART and dynamic programming methods.

Through the HRC of the CART approach via Gini, the result is illustrated as follows:

HRC	Jeffreys	Student	Clopper-Pearson
1	0.0001044	0.0008983	0.0007491
2	0.0001433	0.0010016	0.0012339
3	0.0001788	0.0014247	0.0016532
4	0.0002115	0.0018178	0.0019410
5	0.0002444	0.0018288	0.0024369
6	0.0002566	0.0024585	0.0027762
7	0.0003457	0.0028806	0.0035138
Portfolio	0.0001938	0.0022051	0.0018243

Table 15: MoC Calibration Results based on HRC obtained with CART approach using Jeffreys, Student, and Clopper-Pearson methods.

Using the classes obtained by dynamic programming, we obtain the following MoCs.

HRC	Jeffreys	Student	Clopper-Pearson
1	0.0001783	0.0015160	0.0011372
2	0.0001209	0.0010638	0.0008935
3	0.0001365	0.0013345	0.0011332
4	0.0001545	0.0016423	0.0014162
5	0.0001185	0.0015039	0.0013222
Portfolio	0.0001308	0.0048231	0.0012460

Table 16: MoC Calibration Results based on HRC obtained with dynamic programming approach using Jeffreys, Student, and Clopper-Pearson methods.

The results indicate that the MoC values vary significantly depending on the statistical approach used. The Jeffreys method produces consistently lower MoC values, reflecting its Bayesian nature, which incorporates prior knowledge to smooth estimates. The Student approach yields intermediate values, suggesting a more robust treatment of uncertainty. Finally, the Clopper-Pearson method results in higher MoC values, especially for higher HRC levels, due to its conservative nature as an exact confidence interval method. Comparing the CART and dynamic programming methods, the MoC values obtained with dynamic programming tend to be slightly lower, suggesting that this approach may provide a more efficient estimation framework. However, the choice of method depends on the trade-off between computational efficiency and statistical conservatism in risk assessment. Overall, these findings highlight the importance of selecting an appropriate statistical method based on the desired level of conservatism and the available data characteristics.

The CART approach provides more granular and stable MoC estimates, making it better suited for regulatory compliance and capital allocation. The Dynamic Programming approach, while computationally efficient, may introduce greater uncertainty and model risk due to the creation of large, heterogeneous segments. The choice of approach should depend on the specific requirements of the regulatory environment and the risk profile of the portfolio.

Discussion

As with any study, it is essential to discuss the key findings while highlighting the limitations of the methods used and suggesting potential improvements for more accurate results. This study had a dual objective, consisting of constructing homogeneous risk classes (HRCs) through various segmentation techniques and conducting related tests, and estimating uncertainties around the risk score both overall and within each HRC, while accounting for data errors and estimation uncertainties to avoid underestimating risk. Among the tested approaches, CART decision trees and dynamic programming provided the most stable HRCs in terms of default rate consistency over time. However, they still present some limitations:

- **Instability of certain HRCs over time**: Some risk classes disappear, raising concerns about the consistency and reliability of the segmentation model.
- **Limited number of HRCs**: Only a minimal number of HRCs (5) were retained due to constraints such as the limited data quality of the dataset used.
- **Dynamic programming limitations**: While effective at defining optimal boundaries, this method is highly sensitive to parameters like the crossover probability, which can impact result stability. Additionally, its complexity may limit regulatory acceptance.

Given these factors, **the CART approach appears to be the most suitable** for future use, as it balances **efficiency, stability, and regulatory acceptability**. Once the HRCs were defined, different methodologies were used to estimate the MoC C, including:

- Bootstrap approach
- Parametric approach
- Jeffrey's method
- Student's approach
- Clopper-Pearson method

Table 17 presents the relative differences in MoC C estimates for the HRCs obtained using the CART approach, compared to the most commonly used method, the bootstrap approach. The results show that the parametric,the student and the Clopper-Pearson methods tend to yield higher MoC C estimates for certain HRCs (specifically, classes 1, 2, 4, and 6 for the parametric method, and classes 2, 3, 4, 5, and 6 for Clopper-Pearson), when compared to the bootstrap estimates for these categories. On the other hand, for the Bayesian, Jeffreys, and Student methods, the bootstrap approach provides a more conservative estimate across all HRCs, meaning it produces higher values. In fact, for these three methods, the bootstrap estimate is at least 70% higher for every HRC.

At the portfolio level, the bootstrap approach consistently results in the highest MoC C estimate, except for the Student method. Specifically, bootstrap produces an estimate that is 20.65% higher than the parametric approach, 89.63% higher than the Bayesian method, 91.19% higher than the Jeffreys method and 17.04% higher than the Clopper-Pearson method. The Student method performs slightly better than the bootstrap approach at the portfolio level, as it provides a MoC estimate that is 0.28% higher.
[5]

[5] The relative differences were calculated using the following formula:

$$\Delta\% = 100 \times \frac{\text{Bootstrap} - \text{Method}}{\text{Bootstrap}}$$

where Method refers to the estimate obtained from the given approach.

HRC	Bootstrap	Parametric(%)	Bayesien(%)	Jeffreys(%)	Student(%)	Clopper-Pearson(%)
1	0.000769	-16.51	74.64	86.42	-16.81	2.59
2	0.000974	-2.57	79.43	85.29	-2.83	-26.68
3	0.001648	13.77	86.94	89.15	13.55	-0.32
4	0.001730	-4.80	86.97	87.77	-5.07	-12.20
5	0.002016	9.52	87.43	87.88	9.29	-20.88
6	0.002178	-12.58	87.76	88.22	-12.88	-27.47
7	0.006412	55.19	95.28	94.61	55.07	45.20
Portfolio	0.002199	20.65	89.63	91.19	-0.28	17.04

Table 17: Comparison of relative differences across various MoC estimation methods

For HCRs obtained using the dynamic programming approach, the relative differences compared to the Bootstrap method, are presented in Table 18. The results are less conclusive than those obtained with the HCRs using the CART method. In fact, all other methods provide a better estimate of MoC C for the least risky class compared to the Bootstrap approach. Regarding the portfolio, the Student method provides a slightly better estimate of MoC C than the Bootstrap method. Specifically, the Bootstrap method gives an estimate of MoC C that is 0.67% lower than the one obtained with the Student approach. Furthermore, the Bootstrap method significantly overestimates MoC C, making it 70% higher than the estimate from the parametric method, 96.29% higher than the Bayesian approach, 97.27% higher than the Jeffreys method, and 74% higher than the Clopper-Pearson approach.

HRC	Bootstrap(%)	Parametric(%)	Bayesien(%)	Jeffreys(%)	Student(%)	Clopper-Pearson(%)
1	0.008976	-789.41	-178.29	-4.88	-791.76	-568.94
2	0.024619	-24.24	72.70	85.84	-24.57	-4.63
3	0.050308	-6.14	83.15	89.11	-6.42	9.63
4	0.090695	19.31	90.40	92.39	19.10	30.24
5	0.209576	82.60	98.54	98.63	82.56	84.66
Portfolio	0.128624	69.99	96.29	97.27	-0.67	73.99

Table 18: Comparison of relative differences across various MoC estimation methods for dynamic programming

Although in the literature, other approaches, particularly the Pearson approach, may be more conservative and therefore preferred over the Bootstrap approach, in the context of this study and given the results, overall, the Bootstrap method appears to be more conservative. This result may, however, be limited by the weak quality of the model that was provided.

Figure 4 presents the density curves of scores for defaulting observations, non-defaulting observations, and the overall population. The lack of clear separation between the curves for defaulting and non-defaulting observations suggests that the scores do not effectively discriminate between these groups. This finding underscores a critical limitation of the given built model: **its weak discriminatory power**. A model with strong discriminatory power would show distinct density curves for defaulting and non-defaulting observations, enabling better risk differentiation. This aligns with the earlier observation of a low Gini index (0.48), further highlighting the need for model improvements.

To address this issue, future work could focus on refining the scoring model by revising the modeling step to follow a rigorous methodology that meets regulatory requirements. The way the model is constructed is essential to ensure effective discrimination between defaulting and non-defaulting obligors. This involves adhering to a strict methodology for scoring modeling, which includes:

- **Data Quality and Preprocessing**: Ensuring that the input data is clean, representative, and free from biases. This includes addressing missing values, outliers, and potential data imbalances.

- **Feature Engineering**: Identifying and incorporating predictive variables that better capture the differences between defaulting and non-defaulting obligors. This could involve domain-specific knowledge and advanced techniques such as interaction terms or non-linear transformations.

- **Regular Updates**: Continuously updating the model with new data to ensure it remains relevant and accurate over time, especially in dynamic economic environments.

Additionally, the **number of classes and their distribution** play a crucial role in ensuring meaningful risk differentiation and loss quantification. The segmentation into homogeneous risk classes (HRCs) should strike a balance between having enough classes to capture risk heterogeneity and avoiding excessive concentration of exposures or obligors in a single class. Excessive concentration in a single class may indicate inadequate risk differentiation, while too few exposures or obligors in a class can lead to unreliable estimates of the MoC. In this study, the dynamic programming and CART approaches were retained to define HRCs. However, the instability of certain HRCs over time, as noted earlier, raises concerns about the consistency of the segmentation. To improve this, the distribution of exposures across classes should be monitored to avoid excessive concentration in a single class, unless justified by empirical evidence of risk homogeneity. This ensures that each class represents a distinct risk profile and contributes meaningfully to the overall risk assessment.

By addressing these issues, the model's ability to discriminate between defaulting and non-defaulting obligors can be enhanced, leading to more accurate risk estimates and better-balanced MoC estimation. Furthermore, these improvements will strengthen the model's compliance with regulatory requirements and its overall reliability.

Conclusion

This project explored various approaches for constructing homogeneous risk classes (HRCs) and estimating margins of conservatism (MoC) in the context of credit risk for a mortgage loan portfolio. The primary objective was to ensure compliance with regulatory requirements regarding model quality, particularly the temporal stability of default rates and volumes within each risk class, while maintaining heterogeneity between classes and homogeneity within each class with respect to risk drivers.

The results demonstrate that the CART (supervised) and dynamic programming (unsupervised) segmentation methods are the most effective for constructing HRCs. Both approaches yielded risk classes that exhibited strong stability in terms of default rates and volumes over the period 2011-2017. Specifically, the CART method produced seven risk classes with increasing risk levels, while the dynamic programming method resulted in five risk classes. The heterogeneity of these classes was rigorously validated using appropriate statistical tests, ensuring their reliability for risk management purposes.

For the estimation of MoC, several methods among which the bootstrap method were applied to both sets of HRCs. The results showed that the MoC estimates were more conservative when using the CART-based HRCs compared to those derived from dynamic programming. This suggests that the CART-based segmentation, with its finer granularity and higher number of risk classes, provides a more conservative approach to capital requirement estimation. In a conservative risk management framework, this approach should be prioritized.

However, certain limitations were identified, including the sensitivity of the dynamic programming method to specific parameters, such as the crossover probability, which could affect the stability of the resulting risk classes. These findings highlight the importance of careful parameter calibration and ongoing monitoring of class stability to ensure robust risk classification.

To sum up, this study provides a solid foundation for improving risk classification and MoC estimation in credit risk management. The CART and dynamic programming methods, combined with the bootstrap approach for MoC estimation, offer reliable and conservative solutions for regulatory compliance. Future work should focus on enhancing the stability of risk classes, refining segmentation techniques, and exploring advanced machine learning methods to further improve discriminatory power and predictive accuracy. By addressing these challenges, the proposed methodologies can be strengthened to better support regulatory compliance and risk management practices.

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A. Welch test Results

CHR 1	CHR 2	t-stat	P-value	Interpretation
1	2	-1614.02	0.00	Different
1	3	-2670.24	0.00	Different
1	4	-2612.95	0.00	Different
1	5	-2653.31	0.00	Different
2	3	-1437.82	0.00	Different
2	4	-1757.96	0.00	Different
2	5	-2011.60	0.00	Different
3	4	-979.28	0.00	Different
3	5	-1474.43	0.00	Different
4	5	-688.60	0.00	Different

Table 19: Welch Train

CHR 1	CHR 2	t-stat	P-value	Interpretation
1	2	-932.8	0.00	Different
1	3	-1543.60	0.00	Different
1	4	-1505.34	0.00	Different
1	5	-1533.27	0.00	Different
2	3	-831.18	0.00	Different
2	4	-1011.33	0.00	Different
2	5	-1161.9	0.00	Different
3	4	-562.37	0.00	Different
3	5	-851.1	0.00	Different
4	5	-397.69	0.00	Different

Table 20: Welch Test

B. Chi-square test Results

Variable	Chi-Square Stat	P-value	Interpretation	Summary
grade:A	662376.18	0.00	Different	
grade:B	318796.48	0.00	Different	
home_ownership:ANY	187.71	0.00	Different	
home_ownership:Mortgage	41337.27	0.00	Different	
home_ownership:OTHER	7.31	0.12	Different	
Purpose:car	1376.95	0.00	Different	
Purpose:credit_car	24040.62	0.00	Different	
Purpose:vacation	1.96	0.74	Homogeneous	
addr_state:AK	3.50	0.47	Homogeneous	
Purpose:small_business	3004.98	0.00	Different	
Homogeneous Variables				4
Heterogeneous Variables				164

Table 21: Chi-Square Train

Variable	Chi-Square Stat	P-value	Interpretation	Summary
grade:A	221666.34	0.00	Different	
grade:B	1060080.32	0.00	Different	
home_ownership:ANY	43.93	0.00	Different	
home_ownership:Mortgage	13806.91	0.00	Different	
home_ownership:OTHER	9.96	0.04	Different	
Purpose:car	78.32	0.00	Different	
Purpose:credit_car	7539.91	0.00	Different	
Purpose:vacation	8.78	0.06	Homogeneous	
addr_state:AK	3.06	0.54	Homogeneous	
Purpose:small_business	924.20	0.00	Different	
Homogeneous Variables				8
Heterogeneous Variables				160

Table 22: Chi-Square Test

C. CART HRCs using Gini diversity index

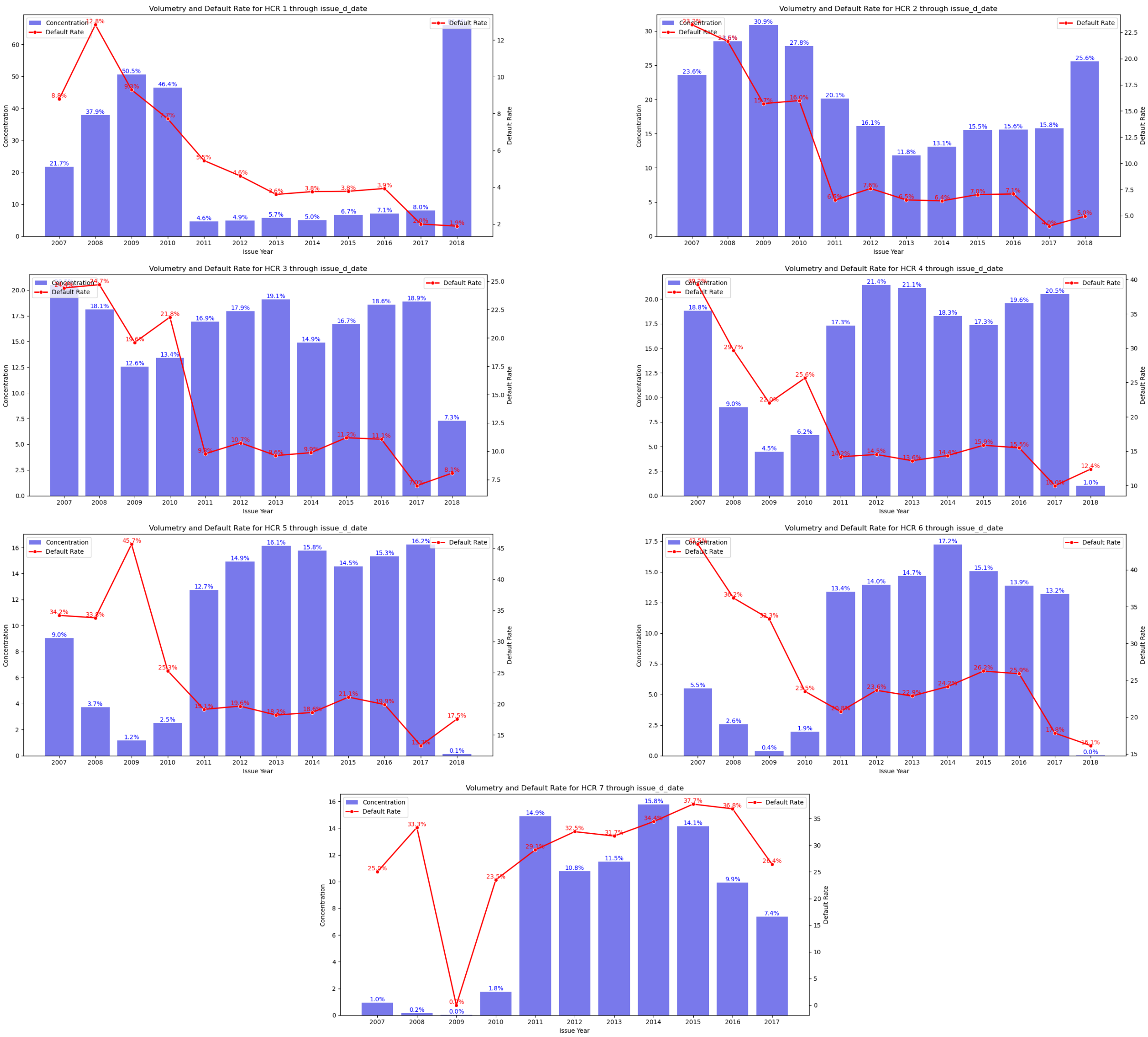

Figure 40: Evolution of concentration and default rate through issue date, for each HCR obtained with CART using Gini diversity index

D. CART HRCs using Shannon Entropy

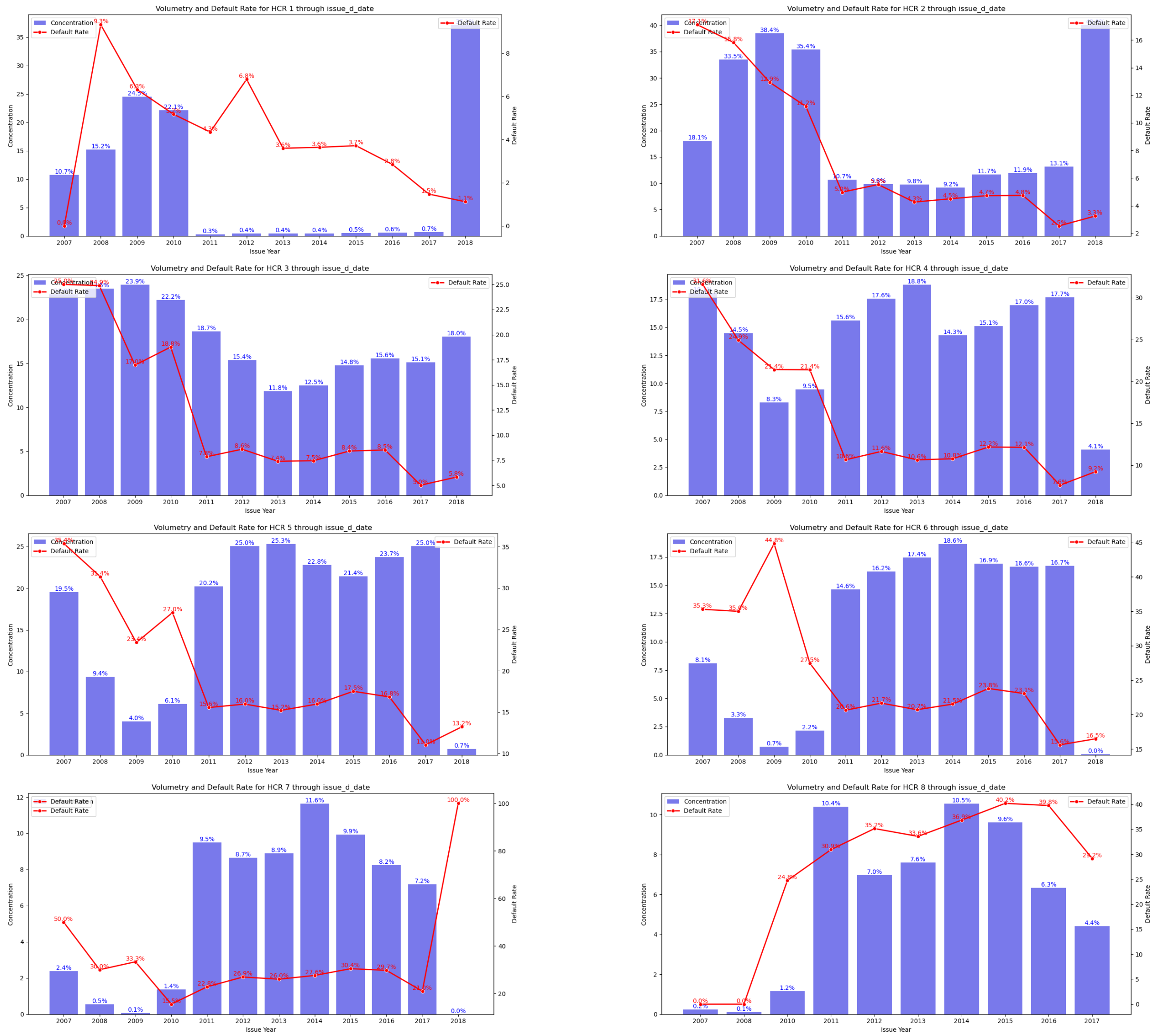

Figure 41: Evolution of concentration and default rate through issue date, for each HCR obtained with CART using Shannon entropy

E. Dynamic programic Approach HRCs

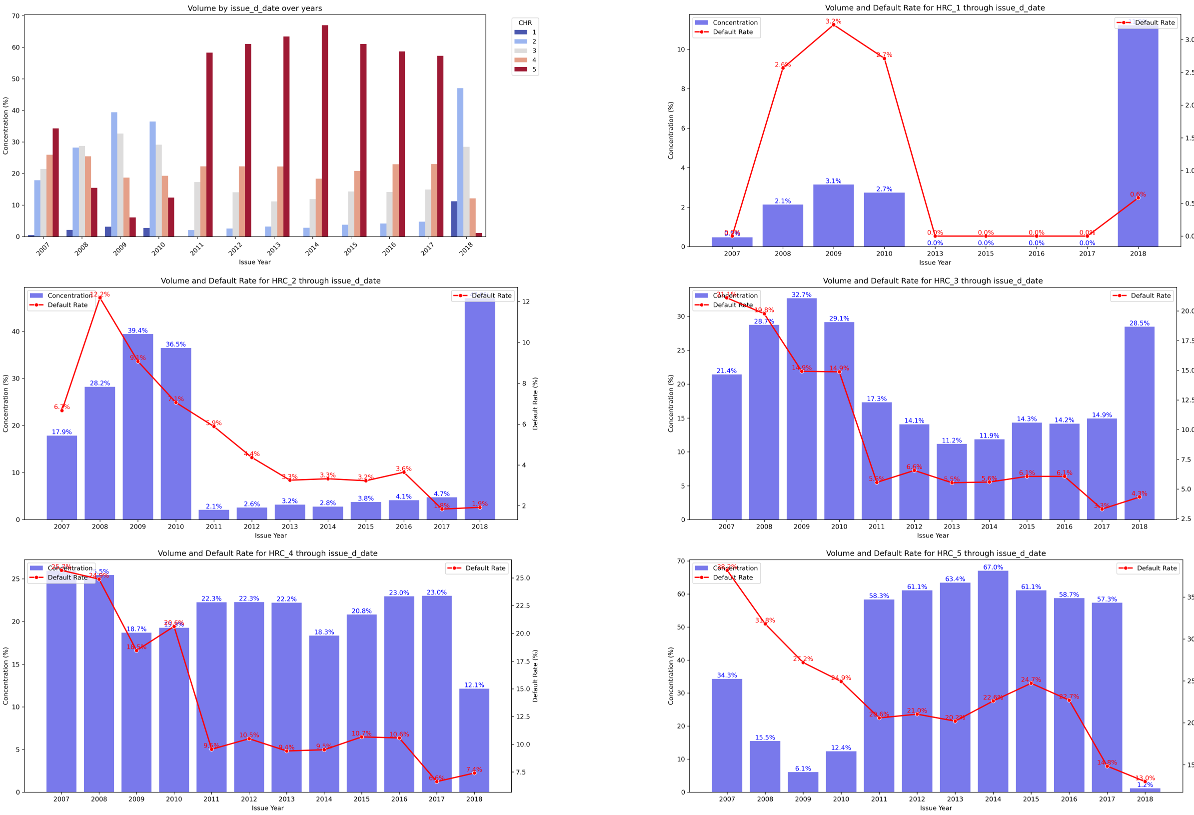

Figure 42: Evolution of concentration and default rate through issue date, for each HCR obtained with Dynamic Programming Approach

F. Quantile Approach HRCs

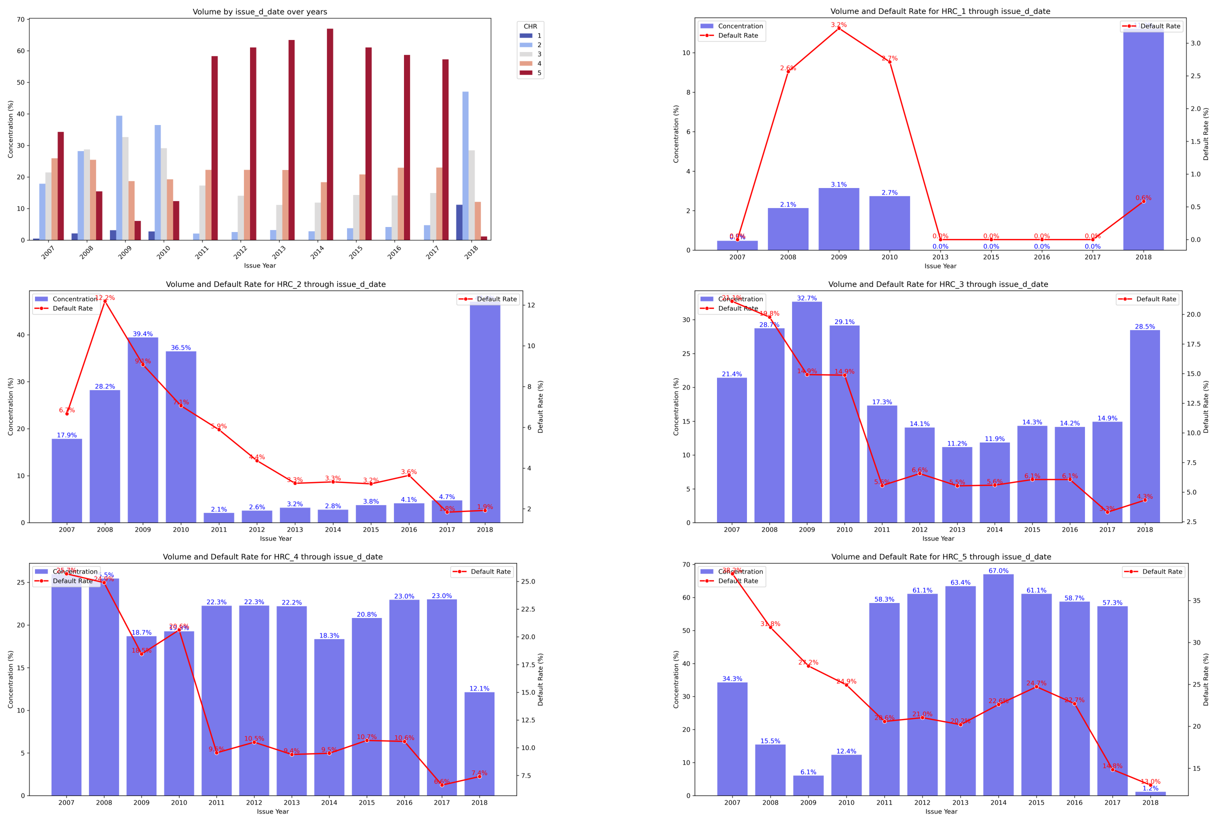

Figure 43: Evolution of concentration and default rate through issue date, for each HCR obtained with Quantile Approach

G. Homogeneity of HRC

Table 23: ANOVA Results by HCR classes obtained with CART Approach using Gini diversity index

Variable	F_statistic	p-value	Decision	CHR
term	0.812685	0.367328	Homogeneous	3
initial_list_status	0.000052	0.994244	Homogeneous	4
acc_now_delinq	0.907107	0.340884	Homogeneous	4
home_ownership	1.084240	0.366667	Homogeneous	4
total_acc	1.084323	0.338132	Homogeneous	4
delinq_2yrs	0.951943	0.385992	Homogeneous	5
pub_rec	2.646454	0.070905	Homogeneous	5
acc_now_delinq	0.317982	0.572824	Homogeneous	5
mths_since_last_delinq	0.386034	0.818795	Homogeneous	5
total_acc	2.914202	0.054250	Homogeneous	5
delinq_2yrs	0.271295	0.762392	Homogeneous	6
pub_rec	2.208473	0.109871	Homogeneous	6
acc_now_delinq	0.002359	0.961264	Homogeneous	6

Table 24: ANOVA Results by HCR classes obtained with CART Approach using Shannon Entropy

Variable	F_statistic	p-value	Decision	CHR
total_acc	2.228774	0.107662	Homogeneous	3
verification_status	2.254751	0.104902	Homogeneous	3
mths_since_last_delinq	2.044691	0.085249	Homogeneous	4
total_acc	2.032870	0.130961	Homogeneous	4
term	0.100817	0.750851	Homogeneous	4
delinq_2yrs	2.101763	0.122243	Homogeneous	4
acc_now_delinq	0.008028	0.928604	Homogeneous	5
acc_now_delinq	0.051067	0.821217	Homogeneous	6
mths_since_last_delinq	0.834340	0.503036	Homogeneous	6
pub_rec	0.075391	0.927380	Homogeneous	6
delinq_2yrs	0.907810	0.403408	Homogeneous	6
mths_since_last_record	1.623549	0.135989	Homogeneous	7
acc_now_delinq	1.102007	0.293828	Homogeneous	7
pub_rec	0.261644	0.769785	Homogeneous	7
delinq_2yrs	0.249792	0.778963	Homogeneous	7
acc_now_delinq	3.212882	0.073064	Homogeneous	8

Table 25: ANOVA Results by HCR classes obtained with Quantile Approach

Variable	F_statistic	p-value	Decision	CHR
acc_now_delinq	NaN	NaN	Homogeneous	1
pub_rec	0.722679	0.485456	Homogeneous	1
Target	2.564660	0.109284	Homogeneous	1
open_acc	0.975636	0.439778	Homogeneous	2
acc_now_delinq	0.058735	0.808508	Homogeneous	2
mths_since_last_delinq	2.246871	0.061431	Homogeneous	2
pub_rec	1.583295	0.205309	Homogeneous	2
delinq_2yrs	2.921166	0.053881	Homogeneous	2
Target	2.730163	0.098476	Homogeneous	2
open_acc	1.949757	0.069070	Homogeneous	3
acc_now_delinq	0.611794	0.434118	Homogeneous	3
total_acc	1.904939	0.148845	Homogeneous	3
verification_status	2.774079	0.062419	Homogeneous	3
mths_since_last_delinq	0.817225	0.513877	Homogeneous	4
pub_rec	1.961696	0.140624	Homogeneous	4
delinq_2yrs	1.614841	0.198927	Homogeneous	4
total_acc	1.392001	0.248579	Homogeneous	5

Table 26: Chi2 test Results by HCR classes obtained with CART using Gini diversity index

Variable	Chi2_stat	df	p-value	Decision	CHR
total_acc	0.967526	2	0.616459	Homogeneous	1
pub_rec	2.798800	2	0.246745	Homogeneous	1
emp_length	3.566328	5	0.613376	Homogeneous	1
open_acc	13.964810	7	0.051811	Homogeneous	2
acc_now_delinq	0.607196	1	0.435846	Homogeneous	2
total_acc	3.460590	2	0.177232	Homogeneous	2
pub_rec	5.623011	2	0.060114	Homogeneous	2
home_ownership	9.431853	5	0.093030	Homogeneous	2
open_acc	11.865558	7	0.105068	Homogeneous	3
pub_rec	0.052827	2	0.973932	Homogeneous	3
verification_status	4.906680	2	0.086006	Homogeneous	3
mths_since_earliest_cr_line	6.343153	5	0.274246	Homogeneous	3
acc_now_delinq	1.510817	1	0.219014	Homogeneous	4
mths_since_last_delinq	7.311367	4	0.120321	Homogeneous	4
total_acc	5.353745	2	0.068778	Homogeneous	4
initial_list_status	0.996492	1	0.318161	Homogeneous	4
pub_rec	0.373273	2	0.829745	Homogeneous	4
term	0.861064	1	0.353441	Homogeneous	4
delinq_2yrs	3.590009	2	0.166127	Homogeneous	4
total_rev_hi_lim	7.499871	7	0.378749	Homogeneous	4
annual_inc	16.082419	11	0.138099	Homogeneous	4
home_ownership	4.046793	5	0.542699	Homogeneous	4
mths_since_last_record	11.605449	6	0.071372	Homogeneous	5
dti	16.453838	9	0.057988	Homogeneous	5
acc_now_delinq	2.254373	1	0.133237	Homogeneous	5
mths_since_last_delinq	3.977849	4	0.409012	Homogeneous	5
total_acc	2.744878	2	0.253488	Homogeneous	5
initial_list_status	2.070016	1	0.150220	Homogeneous	5
pub_rec	2.170445	2	0.337827	Homogeneous	5
term	1.918470	1	0.166025	Homogeneous	5
emp_length	4.052753	5	0.541846	Homogeneous	5
delinq_2yrs	0.045420	2	0.977546	Homogeneous	5
verification_status	4.085376	2	0.129680	Homogeneous	5
total_rev_hi_lim	6.971161	7	0.431889	Homogeneous	5
annual_inc	13.338308	11	0.271790	Homogeneous	5
mths_since_earliest_cr_line	1.539791	5	0.908433	Homogeneous	5
mths_since_last_record	10.974860	6	0.089157	Homogeneous	6
acc_now_delinq	0.223281	1	0.636551	Homogeneous	6
total_acc	0.643333	2	0.724940	Homogeneous	6
pub_rec	0.137044	2	0.933773	Homogeneous	6
emp_length	8.488628	5	0.131283	Homogeneous	6
mths_since_earliest_cr_line	10.465716	5	0.063065	Homogeneous	6
home_ownership	10.792066	5	0.055662	Homogeneous	6
mths_since_last_record	9.980127	6	0.125491	Homogeneous	7
acc_now_delinq	1.867049	1	0.171813	Homogeneous	7
mths_since_last_delinq	2.619449	4	0.623382	Homogeneous	7
total_acc	0.394998	2	0.820781	Homogeneous	7
pub_rec	1.989378	2	0.369838	Homogeneous	7
inq_last_6mths	2.677506	3	0.444064	Homogeneous	7
delinq_2yrs	5.342059	2	0.069181	Homogeneous	7

Table 27: Chi2 test Results by HCR classes obtained with CART using Shannon Entropy

Variable	Chi2_stat	df	p-value	Decision	CHR
acc_now_delinq	0.299290	1	0.584328	Homogeneous	1
total_acc	4.158449	2	0.125027	Homogeneous	1
pub_rec	4.091547	2	0.129280	Homogeneous	1
term	2.002705	1	0.157019	Homogeneous	1
emp_length	4.228240	5	0.517045	Homogeneous	1
home_ownership	4.773482	5	0.444145	Homogeneous	1
acc_now_delinq	1.886440	1	0.169604	Homogeneous	2
emp_length	7.659411	5	0.176032	Homogeneous	2
mths_since_earliest_cr_line	9.054686	5	0.106903	Homogeneous	2
open_acc	13.082783	7	0.070117	Homogeneous	3
acc_now_delinq	0.425307	1	0.514301	Homogeneous	3
total_acc	4.627267	2	0.098901	Homogeneous	3
pub_rec	0.402245	2	0.817812	Homogeneous	3
emp_length	7.349980	5	0.195886	Homogeneous	3
verification_status	5.632601	2	0.059827	Homogeneous	3
mths_since_issue_d	11.249768	7	0.128089	Homogeneous	4
open_acc	12.426424	7	0.087379	Homogeneous	4
initial_list_status	3.633974	1	0.056611	Homogeneous	4
pub_rec	0.536326	2	0.764783	Homogeneous	4
term	1.572128	1	0.209898	Homogeneous	4
total_rev_hi_lim	9.528290	7	0.216917	Homogeneous	4
annual_inc	17.377495	11	0.097195	Homogeneous	4
mths_since_earliest_cr_line	8.739820	5	0.119903	Homogeneous	4
acc_now_delinq	0.217928	1	0.640623	Homogeneous	5
mths_since_last_delinq	4.762368	4	0.312560	Homogeneous	5
initial_list_status	1.781940	1	0.181911	Homogeneous	5
pub_rec	0.546726	2	0.760817	Homogeneous	5
term	0.000000	1	1.000000	Homogeneous	5
emp_length	10.094029	5	0.072614	Homogeneous	5
delinq_2yrs	2.051455	2	0.358536	Homogeneous	5
total_rev_hi_lim	12.848337	7	0.075891	Homogeneous	5
annual_inc	9.780781	11	0.550207	Homogeneous	5
mths_since_earliest_cr_line	7.073924	5	0.215200	Homogeneous	5
home_ownership	5.610298	5	0.346003	Homogeneous	5
mths_since_last_record	11.973093	6	0.062572	Homogeneous	6
open_acc	13.546766	7	0.059855	Homogeneous	6
acc_now_delinq	3.534686	1	0.060098	Homogeneous	6
mths_since_last_delinq	6.218629	4	0.183405	Homogeneous	6
total_acc	0.015094	2	0.992481	Homogeneous	6
initial_list_status	0.224140	1	0.635903	Homogeneous	6
pub_rec	3.795634	2	0.149895	Homogeneous	6
emp_length	4.875111	5	0.431311	Homogeneous	6
delinq_2yrs	2.699439	2	0.259313	Homogeneous	6
verification_status	5.978240	2	0.050332	Homogeneous	6
annual_inc	10.229479	11	0.509871	Homogeneous	6
mths_since_earliest_cr_line	3.362201	5	0.644333	Homogeneous	6
home_ownership	9.791657	5	0.081358	Homogeneous	6
mths_since_last_record	6.195225	6	0.401680	Homogeneous	7
acc_now_delinq	1.014577	1	0.313809	Homogeneous	7
mths_since_last_delinq	7.291030	4	0.121285	Homogeneous	7
total_acc	0.835591	2	0.658497	Homogeneous	7
pub_rec	2.984279	2	0.224891	Homogeneous	7
verification_status	4.121947	2	0.127330	Homogeneous	7

Table 28: Chi2 test Results by HCR classes obtained with Quantile

Variable	Chi2_stat	df	p-value	Decision	CHR
mths_since_issue_d	0.115809	3	0.989875	Homogeneous	1
mths_since_last_record	3.654628	6	0.723296	Homogeneous	1
open_acc	6.859938	6	0.333995	Homogeneous	1
acc_now_delinq	0.000000	0	1.000000	Homogeneous	1
mths_since_last_delinq	4.754987	4	0.313373	Homogeneous	1
total_acc	2.424552	2	0.297519	Homogeneous	1
pub_rec	0.219516	2	0.896051	Homogeneous	1
inq_last_6mths	3.770073	3	0.287386	Homogeneous	1
grade	5.357670	2	0.068643	Homogeneous	1
term	0.196844	1	0.657281	Homogeneous	1
emp_length	5.941845	5	0.311919	Homogeneous	1
total_rev_hi_lim	7.733723	7	0.356659	Homogeneous	1
annual_inc	8.256404	11	0.690168	Homogeneous	1
addr_state	57.342166	49	0.193350	Homogeneous	1
mths_since_earliest_cr_line	10.319058	5	0.066683	Homogeneous	1
home_ownership	6.330353	4	0.175798	Homogeneous	1
mths_since_last_record	4.064251	6	0.667982	Homogeneous	2
acc_now_delinq	0.000000	1	1.000000	Homogeneous	2
mths_since_last_delinq	9.227215	4	0.055664	Homogeneous	2
total_acc	5.498649	2	0.063971	Homogeneous	2
term	2.724775	1	0.098802	Homogeneous	2
emp_length	3.419644	5	0.635580	Homogeneous	2
delinq_2yrs	1.204680	2	0.547529	Homogeneous	2
mths_since_earliest_cr_line	7.997880	5	0.156353	Homogeneous	2
home_ownership	2.792327	4	0.593158	Homogeneous	2
mths_since_last_record	11.158416	6	0.083602	Homogeneous	3
int_rate	6.270551	4	0.179834	Homogeneous	3
acc_now_delinq	0.764398	1	0.381956	Homogeneous	3
total_acc	0.666292	2	0.716665	Homogeneous	3
grade	8.109541	4	0.087647	Homogeneous	3
emp_length	4.655165	5	0.459395	Homogeneous	3
delinq_2yrs	3.743036	2	0.153890	Homogeneous	3
annual_inc	18.395335	11	0.072849	Homogeneous	3
addr_state	56.768601	50	0.237449	Homogeneous	3
acc_now_delinq	0.000000	1	1.000000	Homogeneous	4
mths_since_last_delinq	8.792101	4	0.066511	Homogeneous	4
total_acc	1.239688	2	0.538028	Homogeneous	4
pub_rec	0.932157	2	0.627458	Homogeneous	4
emp_length	4.774446	5	0.444023	Homogeneous	4
mths_since_earliest_cr_line	10.180190	5	0.070288	Homogeneous	4
acc_now_delinq	3.130952	1	0.076819	Homogeneous	5
total_acc	4.958891	2	0.083790	Homogeneous	5
mths_since_earliest_cr_line	10.687500	5	0.057940	Homogeneous	5
acc_now_delinq	1.627560	1	0.202041	Homogeneous	6